

Re-ERP Procurement Module — Requirements & Design (RD)

Version 1.1 — October 2026 · Module code PO · Status: draft for review

Version	Change
1.1	Purchase orders can be created directly, without a requisition (default); requisitions are optional. Supplier data comes only from the existing RR_SUPPLIER_* tables — section 4.10 lists every table and column used.
1.0	First issue

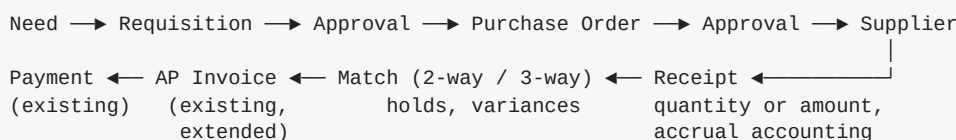
Prepared for the Re-ERP product team. Defines the scope, processes, data model, accounting, services, screens and validations of the native Re-ERP Procurement (Purchasing) module, built primarily for **expense (non-stock) purchasing**: services, consumables, office supplies, IT equipment below the asset threshold, maintenance, rent, professional fees. It plugs into the existing Re-ERP Suppliers, Payables, Tax, Approvals, SLA and General Ledger, and keeps the hooks the Inventory RD (v2.8) reserved for stock purchasing.

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1. Executive Summary

Re-ERP today runs Payables, Receivables, Cash, Fixed Assets, Tax and General Ledger natively. Purchasing is only *viewed* from Oracle Fusion (the `/procurement/*` screens read Fusion REST). This module makes Procure-to-Pay native:



The four design principles:

- Expense first.** Every line is charged to a GL account at order time. There is no on-hand stock; a receipt confirms goods arrived or a service was performed, and (optionally) accrues the liability.
- One quantity ledger.** Each PO distribution carries ordered, received, billed and cancelled quantities and amounts. Receiving and invoicing only move these counters, so open commitments, uninvoiced receipts (GRNI) and PO status are always derivable and always reconcile.
- Reuse, don't rebuild.** Suppliers (RR_SUPPLIER_MASTER / RR_SUPPLIER_SITES), tax codes (RR_INPUT_OUTPUT_TAX), business units (RR_GL_BUSINESS_UNITS), the approval engine (RR_APPROVAL_*, which already lists PROCUREMENT as a module), SLA (RR_SLA_ACCOUNTING_HEADERS/LINES), GL journals and AP invoices (whose lines already have PURCHASE_ORDER_NUMBER, RECEIPT_NUMBER columns) are used as they are.

4. **Same service style as Inventory.** Reads go through the guarded read gateway (`ai/executequery`) on `RR_PO_V_*` views; writes go through one dispatcher endpoint (`po/execute`) that calls whitelisted package procedures. All rules live in PL/SQL packages.

1.1 Functional coverage map

Area	Contents	Section
Setup	BU purchasing options, locations, categories, expense-item catalog, buyers, requester defaults, account derivation, numbering; suppliers from the existing <code>RR_SUPPLIER_*</code> tables (+ optional site options)	4
Requisitions	Catalog and free-text requests, multi-distribution charge accounts, approval, buyer assignment, return to requester	5.1, 6.1
Purchase orders	Direct POs without a requisition (standard path) , quantity and amount (service) lines, schedules, distributions, approval, PDF and email to supplier	5.2, 6.2
Autocreate	Approved requisition lines → POs, grouped by supplier, site, currency and BU	6.3
Change orders	Revisions with re-approval, archive of every approved version	5.3, 6.4
Receiving	Receive by quantity or amount, returns, corrections, receipt accrual	5.4, 6.5
Invoice matching	2-way and 3-way match from AP, tolerances, holds, price and exchange variances	5.5, 6.6
Accruals	Receipt accrual or period-end accrual, uninvoiced-receipts reconciliation, write-off	6.7, 7
Close and cancel	Automatic close for receiving and invoicing, manual close, final close, cancel	6.8
Accounting	Event model, Dr/Cr rules, worked examples, SLA and GL hand-off	7
Security	Roles, BU scoping, segregation of duties	8
Services and UI	Read views, package specifications, menu, screens, wireframes	9, 10
Reports	Open POs, 3-way status, GRNI, spend, aging, approval turnaround	11
Rules	Validation catalog per feature	12
Rollout	Scripts, seeding, migration from Fusion, phases, open decisions	13–15

1.2 Out of scope (later phases or other modules)

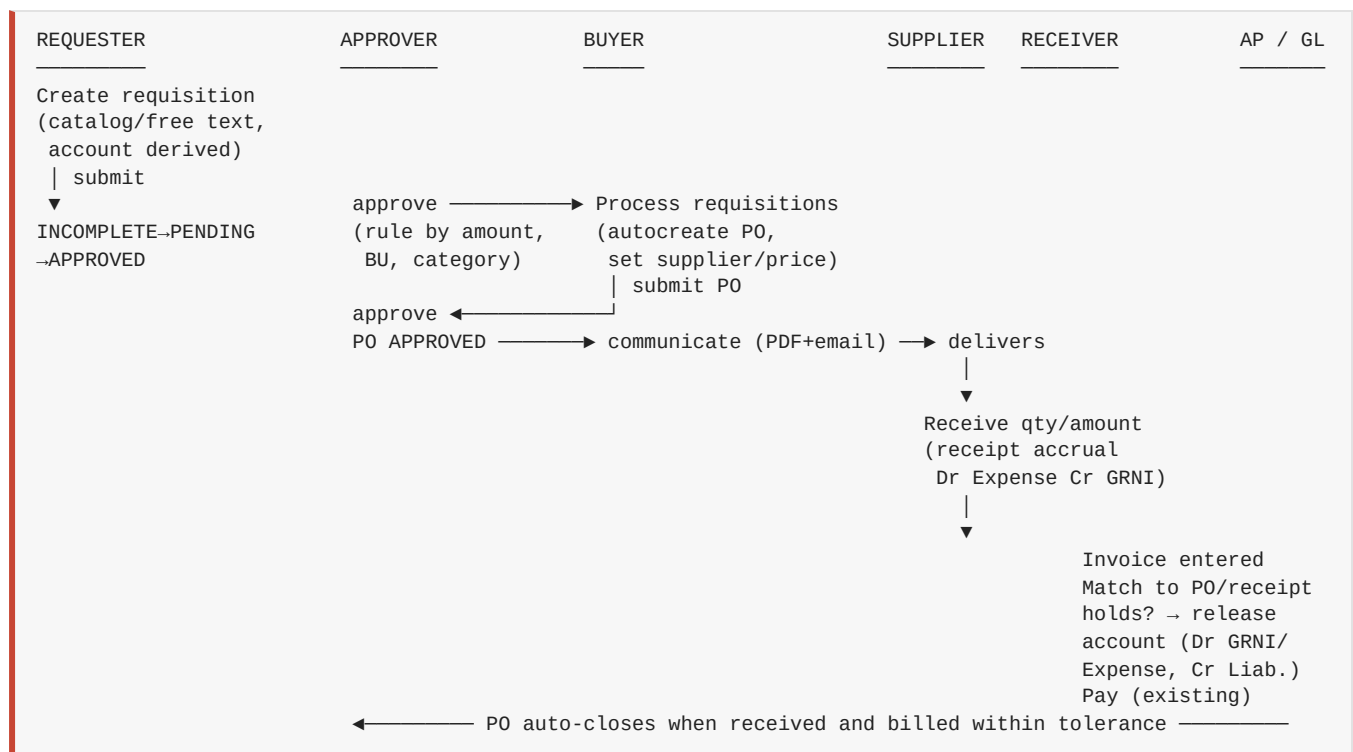
- **Inventory (stock) lines** — line type `INVENTORY` and the `PO_RECEIPT` inventory transaction reserved in the Inventory RD; delivered in Phase 2b once Inventory 1c (transaction engine) is live.
- **Blanket purchase agreements and releases** — Phase 2 (tables named in 5.7 so Phase 1 columns are ready).
- **Budgetary control and encumbrance** — Phase 3 (funds check hooks in 6.9).
- **Sourcing** (RFQs, quotations, bid analysis), **supplier portal**, **punch-out catalogs** — Phase 4.
- **Supplier onboarding** — suppliers stay mastered where they are today (Fusion sync / AP). Procurement only adds purchasing options per supplier site.
- **Capital purchases** — a category can be flagged `CAPEX` ; the AP-to-Fixed-Assets hand-off remains the existing FA additions process.

2. Business Process

2.1 Roles

Role	Does
Requester	Raises requisitions for what they need; receives services/goods delivered to them
Preparer	Raises a requisition on behalf of a requester
Approver	Approves requisitions, POs and change orders (existing approval engine users)
Buyer	Converts requisitions to POs, negotiates price, sends the PO, manages changes and closure
Receiver	Records receipts, returns and corrections (often the requester)
AP clerk	Enters supplier invoices and matches them to POs/receipts; releases holds
Procurement manager	Setup, reports, final close, write-offs

2.2 End-to-end flow



Short paths are supported without bending the model:

- **Direct PO (standard path)** — a buyer creates the PO straight away, without a requisition (section 6.2.0). Requisitions are optional; a BU can make them mandatory with `REQUIRE_REQUISITION = Y`.
- **Two-way services** — an amount line with match level `TWO_WAY` is invoiced directly against the PO; no receipt is needed.
- **Emergency / after-the-fact** — allowed only when the BU option `ALLOW_AFTER_FACT_PO = Y`; such POs are flagged and listed in the compliance report.

2.3 Line types

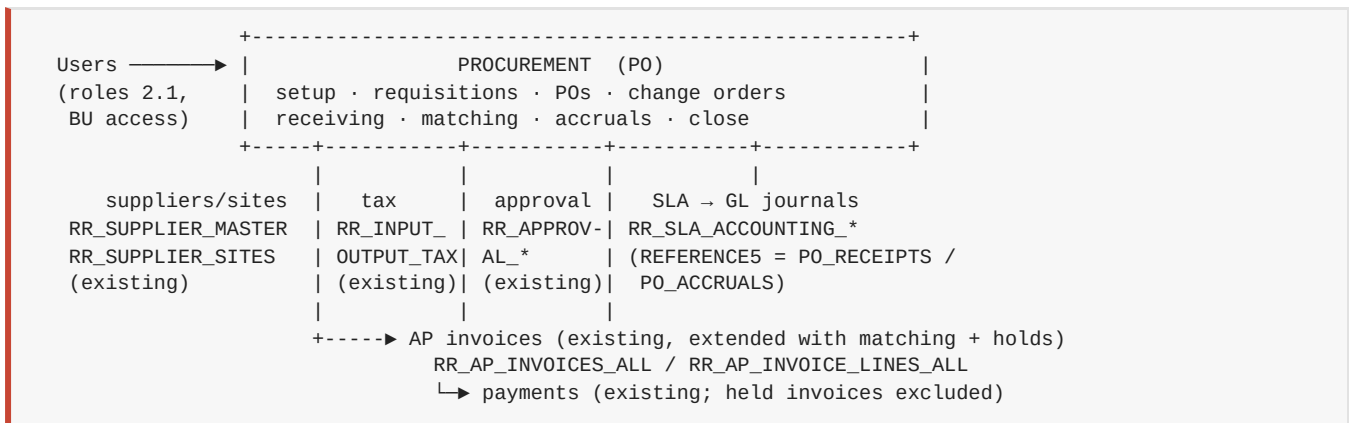
Line type	Ordered as	Received as	Matched as	Typical use
QUANTITY	quantity × unit price, with UOM	quantity	quantity × invoice price	Office chairs, toner, laptops, PPE

Line type	Ordered as	Received as	Matched as	Typical use
AMOUNT	amount only (quantity is 1 and implied)	amount	amount	Consultancy, maintenance, rent, events, milestones

Both line types are expense lines: the charge account is on the distribution. **INVENTORY** (Phase 2b) will add a destination type of inventory and route the receipt to the Inventory engine.

3. Architecture and Conventions

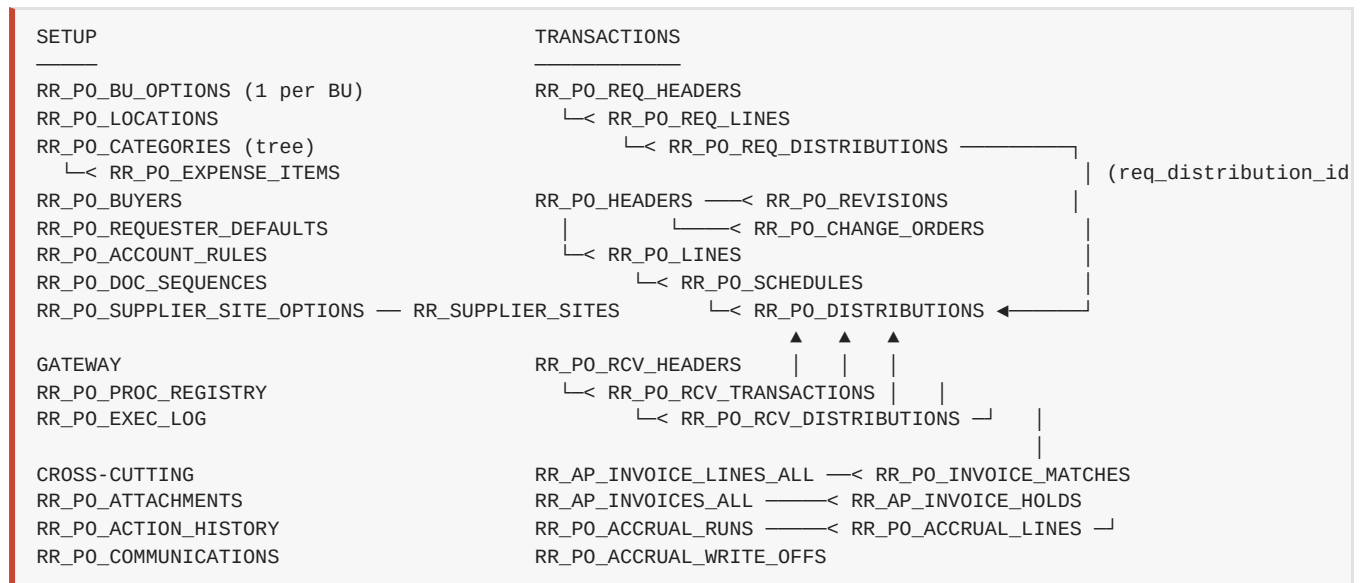
3.1 Where the module sits



3.2 Conventions (same as the Inventory RD)

- **Tables** are prefixed `RR_PO_`; one AP-side table, `RR_AP_INVOICE_HOLDS`, is added to Payables.
- **Primary keys** come from one sequence `RR_PO_SEQ` starting at 6000000001 (Inventory uses 5000000001).
- **WHO columns** on every table: `CREATED_BY VARCHAR2(150)`, `CREATION_DATE TIMESTAMP DEFAULT SYSTIMESTAMP`, `LAST_UPDATED_BY VARCHAR2(150)`, `LAST_UPDATE_DATE TIMESTAMP`, `LAST_UPDATE_LOGIN VARCHAR2(100)`. Stamped by the packages from `p_user`. Immutable tables (receipt transactions, matches, revisions, accrual lines, action history, execution log) are insert-only. Where a table below says *Audit columns* it means exactly this set.
- **Business unit** is always `BUSINESS_UNIT_ID` (the Fusion BU id kept in `RR_GL_BUSINESS_UNITS`), never the name.
- **Money**: every document carries its transaction currency plus `RATE_TYPE`, `RATE_DATE`, `RATE` (from `RR_CURRENCY_DAILY_RATES`). Functional amounts are computed with the frozen rate and rounded to 2 decimals.
- **Accounts** are full code combinations as strings (01-10-200-6105001-0000-000-00-000-000), validated against the GL combinations table, the same way AP stores `DISTRIBUTION_COMBINATION`.
- **Dates** are `DATE` (no time) for business dates; `TIMESTAMP` for audit.
- **Soft rules, hard data**: nothing is physically deleted after approval. Drafts can be deleted; everything else is cancelled or closed.

3.3 Entity-relationship overview



3.4 Complete database object inventory

Object	Kind	Phase	Purpose
RR_PO_SEQ	sequence	1a	All primary keys
RR_PO_BU_OPTIONS	table	1a	Purchasing options per business unit
RR_PO_LOCATIONS	table	1a	Ship-to, bill-to and deliver-to addresses
RR_PO_CATEGORIES	table	1a	Purchasing category tree with default expense account
RR_PO_EXPENSE_ITEMS	table	1a	Expense-item catalog (non-stock)
RR_PO_UOMS	table	1a	Units of measure (shared definition with RR_INV_UOMS , see 4.5)
RR_PO_BUYERS	table	1a	Who can buy, for which BU / category
RR_PO_REQUESTER_DEFAULTS	table	1a	Requester's default BU, deliver-to and charge account
RR_PO_ACCOUNT_RULES	table	1a	Segment overrides for charge-account derivation
RR_PO_DOC_SEQUENCES	table	1a	Document numbering per BU and document type
RR_PO_SUPPLIER_SITE_OPTIONS	table	1a	Purchasing options per supplier site
RR_PO_REQ_HEADERS / _LINES / _DISTRIBUTIONS	tables	1b	Requisitions
RR_PO_HEADERS / _LINES / _SCHEDULES / _DISTRIBUTIONS	tables	1c	Purchase orders
RR_PO_REVISIONS	table	1c	JSON snapshot of every approved PO revision
RR_PO_CHANGE_ORDERS	table	1g	Proposed changes awaiting approval
RR_PO_RCV_HEADERS / _TRANSACTIONS / _DISTRIBUTIONS	tables	1d	Receipts, returns, corrections
RR_PO_INVOICE_MATCHES	table	1e	Invoice line ↔ PO distribution / receipt links
RR_AP_INVOICE_HOLDS	table	1e	Matching holds on AP invoices (Payables-owned)

Object	Kind	Phase	Purpose
RR_PO_ACCRUAL_RUNS / _LINES	tables	1f	Period-end accrual runs
RR_PO_ACCRUAL_WRITE_OFFS	table	1f	Write-off of old uninvoiced receipt accruals
RR_PO_ATTACHMENTS	table	1b	Files on requisitions, POs, receipts
RR_PO_ACTION_HISTORY	table	1b	Status changes and actions on every document
RR_PO_COMMUNICATIONS	table	1c	PO sent to supplier log
RR_PO_PROC_REGISTRY / RR_PO_EXEC_LOG	tables	1a	Write-gateway whitelist and call log
RR_PO_AGREEMENTS / _LINES	tables	2	Blanket agreements (reserved)
RR_PO_BUDGETS / RR_PO_FUNDS_LEDGER	tables	3	Budgetary control (reserved)
RR_PO_V_* (section 9.1)	views	1a–1f	One read view per screen
RR_PO_*_PKG (section 9.3)	packages	1a–1g	All business logic
RR_APPROVAL_CALLBACKS	table	1b	Approval-engine extension: who to call on a decision
RR_APPROVAL_RULES + BUSINESS_UNIT_ID , CATEGORY_CODE	columns	1b	Approval-engine extension: rule filters

4. Setup Data Model

4.1 RR_PO_BU_OPTIONS — purchasing options per business unit (1 row per BU)

Everything a BU decides once: currency defaults, match level, tolerances, accrual method, accounts, approvals and numbering behaviour. Values that drive accounting or matching are **copied onto the PO schedule at approval**, so changing an option later never changes the meaning of an approved PO.

Column	Type	Req	Description
BU_OPTION_ID	NUMBER	Y	PK
BUSINESS_UNIT_ID	NUMBER	Y	RR_GL_BUSINESS_UNITS.BUSINESS_UNIT_ID , unique
FUNCTIONAL_CURRENCY	VARCHAR2(15)	Y	From the BU's primary ledger (e.g. AED)
DEFAULT_RATE_TYPE	VARCHAR2(30)	Y	Default Corporate
DEFAULT_SHIP_TO_LOCATION_ID	NUMBER	N	RR_PO_LOCATIONS
DEFAULT_BILL_TO_LOCATION_ID	NUMBER	N	RR_PO_LOCATIONS
MATCH_LEVEL_QUANTITY	VARCHAR2(10)	Y	TWO_WAY / THREE_WAY for quantity lines (default THREE_WAY)
MATCH_LEVEL_AMOUNT	VARCHAR2(10)	Y	Same for amount (service) lines (default TWO_WAY)
ACCUE_AT_RECEIPT_FLAG	VARCHAR2(1)	Y	Y = receipt accrual; N = period-end accrual (see 7.1)
RECEIPT_ACCRUAL_ACCOUNT	VARCHAR2(200)	Y	GRNI / accrued liabilities combination
PRICE_VARIANCE_ACCOUNT	VARCHAR2(200)	N	IPV account; NULL = post variance to the charge account (usual for expense)
EXCHANGE_GAIN_ACCOUNT	VARCHAR2(200)	Y	Realised ERV gain on match
EXCHANGE_LOSS_ACCOUNT	VARCHAR2(200)	Y	Realised ERV loss on match
ACCRUAL_WRITE_OFF_ACCOUNT	VARCHAR2(200)	N	Default account for write-offs (6.7.3)

Column	Type	Req	Description
ACCRUAL_WRITE_OFF_AGE_DAYS	NUMBER	N	Open distributions with receipt accruals older than this may be written off (NULL = closed distributions only)
SOD_BUYER_RECEIVE	VARCHAR2(1)	Y	N = the PO's buyer may not receive against it (default Y)
OVER_RECEIPT_TOLERANCE_PCT	NUMBER(5,2)	Y	Default 0
OVER_RECEIPT_ACTION	VARCHAR2(10)	Y	REJECT / WARNING
EARLY_RECEIPT_DAYS	NUMBER	N	Receipt allowed this many days before need-by (NULL = no check)
INVOICE_QTY_TOLERANCE_PCT	NUMBER(5,2)	Y	Billed vs ordered/received (default 0)
INVOICE_PRICE_TOLERANCE_PCT	NUMBER(5,2)	Y	Invoice price vs PO price (default 0)
INVOICE_AMOUNT_TOLERANCE	NUMBER	N	Absolute amount tolerance in functional currency
RECEIPT_CLOSE_TOLERANCE_PCT	NUMBER(5,2)	Y	Auto closed for receiving when received $\geq$ ordered $\times$ (1 - tol)
INVOICE_CLOSE_TOLERANCE_PCT	NUMBER(5,2)	Y	Auto closed for invoicing when billed $\geq$ basis $\times$ (1 - tol)
REQ_APPROVAL_REQUIRED	VARCHAR2(1)	Y	Default Y
PO_APPROVAL_REQUIRED	VARCHAR2(1)	Y	Default Y
CO_REAPPROVAL_THRESHOLD_PCT	NUMBER(5,2)	Y	Change orders that raise the PO total by more than this need approval (default 0 = always)
AUTOCREATE_MODE	VARCHAR2(10)	Y	MANUAL (buyer workbench) / AUTO (approved catalog lines with a preferred supplier become POs automatically)
REQUIRE_REQUISITION	VARCHAR2(1)	Y	Default N : POs can be created directly. Y = POs only from requisitions (except buyers with DIRECT_PO_ALLOWED)
ALLOW_AFTER_FACT_PO	VARCHAR2(1)	Y	Allow PO dated after the invoice / receipt (flagged)
BUDGET_CONTROL_LEVEL	VARCHAR2(10)	Y	NONE (Phase 1) / ADVISORY / ABSOLUTE (Phase 3)
PO_TERMS_TEXT	CLOB	N	Standard terms and conditions printed on the PO
PO_EMAIL_SUBJECT / PO_EMAIL_BODY	VARCHAR2(400) / CLOB	N	Templates for sending POs (placeholders {PO_NUMBER} , {SUPPLIER})
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	-

4.2 RR_PO_LOCATIONS

Column	Type	Req	Description
LOCATION_ID	NUMBER	Y	PK
LOCATION_CODE	VARCHAR2(30)	Y	Unique (e.g. DIFC-HQ)
LOCATION_NAME	VARCHAR2(240)	Y	
BUSINESS_UNIT_ID	NUMBER	N	NULL = usable by every BU
ADDRESS_LINE1..3, CITY, REGION, COUNTRY, PO_BOX	VARCHAR2	N	Printed on the PO
CONTACT_NAME, PHONE, EMAIL	VARCHAR2	N	Delivery contact
SHIP_TO_FLAG / BILL_TO_FLAG / DELIVER_TO_FLAG	VARCHAR2(1)	Y	Where the location may be used
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE

Column	Type	Req	Description
Audit columns	-	Y	-

4.3 RR_PO_CATEGORIES — purchasing categories

Categories classify spend and **drive the natural account**. A requester picks "IT Accessories" and never types an account; the category supplies segment 4 (natural account).

Column	Type	Req	Description
CATEGORY_ID	NUMBER	Y	PK
CATEGORY_CODE	VARCHAR2(40)	Y	Unique (e.g. IT.ACCESSORIES)
CATEGORY_NAME	VARCHAR2(240)	Y	
PARENT_CATEGORY_ID	NUMBER	N	Tree (2–3 levels typical: IT > IT.ACCESSORIES)
DEFAULT_NATURAL_ACCOUNT	VARCHAR2(30)	N	Segment-4 value; inherited from the parent when NULL
DEFAULT_LINE_TYPE	VARCHAR2(10)	Y	QUANTITY / AMOUNT
DEFAULT_UOM	VARCHAR2(10)	N	For quantity lines
DEFAULT_TAX_CODE	VARCHAR2(50)	N	RR_INPUT_OUTPUT_TAX.TAX_CODE (input tax)
RECEIPT_REQUIRED_FLAG	VARCHAR2(1)	Y	Y forces THREE_WAY on its lines regardless of BU default
CAPEX_FLAG	VARCHAR2(1)	Y	Asset-type spend; lines are flagged for the FA additions process
REQUESTABLE_FLAG	VARCHAR2(1)	Y	Visible to requesters (parents often N)
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	-

4.4 RR_PO_EXPENSE_ITEMS — expense-item catalog

A light catalog of things the company buys repeatedly. Not stock items: no on-hand, no cost. A requester can always order free text instead; catalog items just pre-fill description, category, UOM, price and supplier.

Column	Type	Req	Description
EXPENSE_ITEM_ID	NUMBER	Y	PK
ITEM_CODE	VARCHAR2(40)	Y	Unique (e.g. EXP-TONER-HP26A)
DESCRIPTION	VARCHAR2(240)	Y	
LONG_DESCRIPTION	VARCHAR2(4000)	N	
CATEGORY_ID	NUMBER	Y	RR_PO_CATEGORIES
LINE_TYPE	VARCHAR2(10)	Y	From the category, overridable
UOM_CODE	VARCHAR2(10)	N	Required for quantity items
LIST_PRICE / CURRENCY_CODE	NUMBER / VARCHAR2(15)	N	Indicative price
PREFERRED_SUPPLIER_ID / PREFERRED_SUPPLIER_SITE_ID	NUMBER	N	Used by autocreate
SUPPLIER_ITEM_NUM	VARCHAR2(60)	N	Supplier's part number
LEAD_TIME_DAYS	NUMBER	N	Need-by default = today + lead time
NATURAL_ACCOUNT_OVERRIDE	VARCHAR2(30)	N	Overrides the category's account
TAX_CODE	VARCHAR2(50)	N	Overrides the category's tax code

Column	Type	Req	Description
INV_ITEM_ID	NUMBER	N	Reserved: link to RR_INV_ITEMS when the item becomes stocked (Phase 2b)
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	-

4.5 RR_PO_UOMS — units of measure

Same columns as RR_INV_UOMS in the Inventory RD 4.7 (UOM_CODE, UOM_NAME, UOM_CLASS, BASE_UOM_FLAG, CONVERSION_TO_BASE, STATUS). Script 301 creates RR_PO_UOMS only if RR_INV_UOMS does not exist, and creates a synonym in the other direction otherwise, so both modules always read one list. Seeded: EA, BOX, PK, SET, HR, DAY, MON, YR, KG, LTR, M, LOT.

4.6 RR_PO_BUYERS

Column	Type	Req	Description
BUYER_ID	NUMBER	Y	PK
USER_NAME	VARCHAR2(150)	Y	Re-ERP login
BUSINESS_UNIT_ID	NUMBER	N	NULL = all BUs the user has access to
CATEGORY_ID	NUMBER	N	Specialisation used by buyer assignment (6.1.4)
DIRECT_PO_ALLOWED	VARCHAR2(1)	Y	Default Y. Only matters when the BU sets REQUIRE_REQUISITION = Y: buyers with Y may still create direct POs
EMAIL / PHONE	VARCHAR2	N	Printed on the PO as the contact
DEFAULT_FLAG	VARCHAR2(1)	Y	Fallback buyer for the BU
STATUS	VARCHAR2(10)	Y	ACTIVE / INACTIVE
Audit columns	-	Y	-

4.7 RR_PO_REQUESTER_DEFAULTS

Column	Type	Req	Description
DEFAULT_ID	NUMBER	Y	PK
USER_NAME	VARCHAR2(150)	Y	Unique with BUSINESS_UNIT_ID
BUSINESS_UNIT_ID	NUMBER	Y	Default BU for the user's requisitions
DELIVER_TO_LOCATION_ID	NUMBER	N	
CHARGE_ACCOUNT_TEMPLATE	VARCHAR2(200)	Y	Full combination giving the user's cost centre segments; segment 4 is replaced by the category's account (6.1.3)
MANAGER_USER_NAME	VARCHAR2(150)	N	Reserved for supervisor-chain approvals (open decision D3)
Audit columns	-	Y	-

4.8 RR_PO_ACCOUNT_RULES — charge-account derivation overrides

Optional rules applied after the default derivation, in PRIORITY order; the first matching rule per segment wins.

Column	Type	Req	Description
RULE_ID	NUMBER	Y	PK
BUSINESS_UNIT_ID	NUMBER	N	NULL = any BU

Column	Type	Req	Description
CATEGORY_ID	NUMBER	N	NULL = any category (child categories match their parent's rule)
EXPENSE_ITEM_ID	NUMBER	N	NULL = any item
SEGMENT_NUM	NUMBER(2)	Y	1–10
SEGMENT_VALUE	VARCHAR2(30)	Y	Value forced into that segment
PRIORITY	NUMBER	Y	Lower runs first
STATUS	VARCHAR2(10)	Y	
Audit columns	-	Y	-

Example: "all FACILITIES.* spend in BU DIFC goes to cost centre 900" → SEGMENT_NUM = 3, SEGMENT_VALUE = '900'.

4.9 RR_PO_DOC_SEQUENCES — document numbering

Column	Type	Req	Description
DOC_SEQ_ID	NUMBER	Y	PK
BUSINESS_UNIT_ID	NUMBER	Y	
DOC_TYPE	VARCHAR2(10)	Y	REQ, PO, RCV, CO, AGR
PREFIX	VARCHAR2(20)	N	e.g. PO-DIFC-
YEAR_IN_NUMBER	VARCHAR2(1)	Y	Y gives PO-DIFC-2026-00017
NEXT_NUMBER	NUMBER	Y	Incremented with SELECT ... FOR UPDATE inside the creating transaction (gap-free per committed document)
PAD_LENGTH	NUMBER	Y	Default 5
RESET_YEARLY	VARCHAR2(1)	Y	Restart at 1 each fiscal year
CURRENT_YEAR	NUMBER(4)	N	Year the counter belongs to
Audit columns	-	Y	-

Numbers are assigned **on creation** for POs, receipts and change orders, and **on submit** for requisitions (drafts show Draft #<id>), so abandoned drafts never burn numbers.

4.10 Suppliers — existing RR_SUPPLIER_* tables

Procurement has **no supplier master of its own**. Every supplier, site and address on a PO comes from the existing Re-ERP supplier tables, which Payables already uses and the current supplier sync keeps up to date. Procurement only reads them; the one extension (RR_PO_SUPPLIER_SITE_OPTIONS, below) is a 1:1 add-on per site and holds no supplier data.

Existing table	Used for	Columns used
RR_SUPPLIER_MASTER	Supplier LOV, status, tax registration on the PO print	SUPPLIER_ID, SUPPLIER, SUPPLIER_NUMBER, ALTERNATE_NAME, STATUS, INACTIVE_DATE, TAX_REGISTRATION_NUMBER, TAXPAYER_ID, ONE_TIME_SUPPLIER_FLAG
RR_SUPPLIER_SITES	Site LOV, purchasing flags, payment terms default, procurement BU	SUPPLIER_SITE_ID, SUPPLIER_ID, SUPPLIER_SITE, SUPPLIER_SITE_CODE, PROCUREMENT_BU_ID, PURCHASING_FLAG, PAY_SITE_FLAG, PAYMENT_TERMS, PAYMENT_TERMS_ID, ADDRESS_NAME, inactive date / status

Existing table	Used for	Columns used
RR_SUPPLIER_ADDRESS	Address printed on the PO (ordering purpose) and shown on the PO header	SUPPLIER_ID + ADDRESS_NAME (join to the site), ADDRESS_LINE1..4, CITY, STATE, COUNTRY, POSTAL_CODE, FORMATTED_ADDRESS, ADDR_PURPOSE_ORDERING, phone and fax
RR_SUPPLIER_SITE_ASSIGNMENTS	Which business units may use the site; bill-to BU; default ship-to / bill-to locations	SUPPLIER_SITE_ID, CLIENT_BU_ID, BILL_TO_BU_ID, SHIP_TO_LOCATION_CODE, BILL_TO_LOCATION_CODE, STATUS, INACTIVE_DATE

Rules that use them (PO-02, MAT-01):

- Supplier STATUS active and INACTIVE_DATE empty or in the future.
- Site active, PURCHASING_FLAG = 'Y', and usable by the PO's BU: an active RR_SUPPLIER_SITE_ASSIGNMENTS row with CLIENT_BU_ID = PO.BUSINESS_UNIT_ID, or RR_SUPPLIER_SITES.PROCUREMENT_BU_ID = PO.BUSINESS_UNIT_ID.
- The supplier LOV on the PO shows only suppliers with at least one such site; picking the supplier filters the sites; picking the site defaults payment terms (site) and ship-to / bill-to (assignment codes matched to RR_PO_LOCATIONS.LOCATION_CODE, else the BU defaults).
- AP matching uses the same site (or another pay site of the same supplier), so the PO, the invoice and the payment all point at one supplier record.
- Supplier creation and maintenance stay where they are today; a supplier added there is immediately available to Procurement.

RR_PO_SUPPLIER_SITE_OPTIONS — purchasing options per supplier site (extension)

Holds only purchasing settings that the supplier tables do not have. Optional: a site with no row uses the BU defaults. Any NULL column also falls back to the BU option.

Column	Type	Req	Description
SITE_OPTION_ID	NUMBER	Y	PK
SUPPLIER_SITE_ID	NUMBER	Y	Unique; RR_SUPPLIER_SITES.SUPPLIER_SITE_ID
PO_COMMUNICATION	VARCHAR2(10)	Y	EMAIL / PRINT / NONE
PO_EMAIL	VARCHAR2(300)	N	Where POs are sent
DEFAULT_CURRENCY	VARCHAR2(15)	N	
MATCH_LEVEL	VARCHAR2(10)	N	Overrides BU default
INVOICE_QTY_TOLERANCE_PCT / INVOICE_PRICE_TOLERANCE_PCT	NUMBER(5,2)	N	Overrides
PURCHASING_HOLD_FLAG	VARCHAR2(1)	Y	Y blocks new POs to the site (e.g. compliance issue)
HOLD_REASON	VARCHAR2(400)	N	
Audit columns	-	Y	-

Payment terms on the PO default from RR_SUPPLIER_SITES.PAYMENT_TERMS / PAYMENT_TERMS_ID.

5. Transaction Data Model

5.1 Requisitions

A requisition is an **internal request**; it has no supplier commitment and no accounting.

RR_PO_REQ_HEADERS

Column	Type	Req	Description
REQ_HEADER_ID	NUMBER	Y	PK
REQ_NUMBER	VARCHAR2(40)	N	Assigned on submit; unique per BU
BUSINESS_UNIT_ID	NUMBER	Y	
DESCRIPTION	VARCHAR2(240)	Y	
JUSTIFICATION	VARCHAR2(2000)	N	Shown to approvers
PREPARER_USER	VARCHAR2(150)	Y	Who keyed it
URGENT_FLAG	VARCHAR2(1)	Y	
STATUS	VARCHAR2(20)	Y	6.1.1
FUNCTIONAL_CURRENCY	VARCHAR2(15)	Y	From BU options
TOTAL_AMOUNT_FUNC	NUMBER	Y	Maintained by the package (sum of line functional amounts)
APPROVAL_REQUEST_ID	NUMBER	N	RR_APPROVAL_REQUESTS.REQUEST_ID of the current cycle
SUBMITTED_DATE / APPROVED_DATE	DATE	N	
FUNDS_STATUS	VARCHAR2(10)	N	Phase 3 (PASSED , ADVISORY , FAILED)
Audit columns	-	Y	-

RR_PO_REQ_LINES

Column	Type	Req	Description
REQ_LINE_ID	NUMBER	Y	PK
REQ_HEADER_ID	NUMBER	Y	FK
LINE_NUM	NUMBER	Y	Unique within header
LINE_TYPE	VARCHAR2(10)	Y	QUANTITY / AMOUNT
EXPENSE_ITEM_ID	NUMBER	N	Catalog item; NULL for free text
CATEGORY_ID	NUMBER	Y	
ITEM_DESCRIPTION	VARCHAR2(240)	Y	
UOM_CODE	VARCHAR2(10)	Q	Required for quantity lines
QUANTITY	NUMBER	Q	> 0 for quantity lines; 1 for amount lines
UNIT_PRICE	NUMBER	Q	Estimated price in line currency
AMOUNT	NUMBER	Y	Quantity × price, or the entered amount
CURRENCY_CODE, RATE_TYPE, RATE_DATE, RATE	-	Y	Rate 1 for functional currency

Column	Type	Req	Description
AMOUNT_FUNC	NUMBER	Y	AMOUNT × RATE
TAX_CODE	VARCHAR2(50)	N	Estimated tax (informational on the requisition)
NEED_BY_DATE	DATE	Y	≥ today
DELIVER_TO_LOCATION_ID	NUMBER	Y	
REQUESTER_USER	VARCHAR2(150)	Y	Defaults to the preparer
SUGGESTED_SUPPLIER_ID / SUGGESTED_SUPPLIER_SITE_ID	NUMBER	N	
SUGGESTED_SUPPLIER_NAME	VARCHAR2(360)	N	Free text when the supplier is not set up
SUPPLIER_ITEM_NUM	VARCHAR2(60)	N	
NOTE_TO_BUYER	VARCHAR2(1000)	N	
BUYER_ID	NUMBER	N	Assigned on approval (6.1.4)
LINE_STATUS	VARCHAR2(15)	Y	OPEN , ON_PO , RETURNED , CANCELLED
PO_LINE_ID	NUMBER	N	Set when the line is placed on a PO
AGREEMENT_LINE_ID	NUMBER	N	Phase 2 (sourced from a blanket agreement)
Audit columns	-	Y	-

RR_PO_REQ_DISTRIBUTIONS — where the cost will be charged. One per line by default; split by percentage when a cost is shared between cost centres.

Column	Type	Req	Description
REQ_DISTRIBUTION_ID	NUMBER	Y	PK
REQ_LINE_ID	NUMBER	Y	FK
DIST_NUM	NUMBER	Y	
PERCENT	NUMBER(7,4)	Y	Sum per line = 100
QUANTITY / AMOUNT	NUMBER	Y	Line value × percent (last distribution absorbs rounding)
CHARGE_ACCOUNT	VARCHAR2(200)	Y	Derived (6.1.3), editable when the user has the EDIT_ACCOUNT privilege
BUDGET_DATE	DATE	Y	Defaults to need-by date; used by Phase 3 funds check
FUNDS_RESERVED_AMOUNT	NUMBER	N	Phase 3
Audit columns	-	Y	-

5.2 Purchase orders

A PO has four levels. Each level answers one question:

Level	Table	Answers
Header	RR_PO_HEADERS	Who are we buying from, in what currency, on what terms?
Line	RR_PO_LINES	What are we buying and at what price?
Schedule	RR_PO_SCHEDULES	When and where is it delivered, and how is it matched? (one line can be delivered in several schedules, e.g. monthly service)

Level	Table	Answers
Distribution	RR_PO_DISTRIBUTIONS	Which account pays for it? Holds the quantity ledger (ordered / delivered / billed / cancelled).

RR_PO_HEADERS

Column	Type	Req	Description
PO_HEADER_ID	NUMBER	Y	PK
PO_NUMBER	VARCHAR2(40)	Y	Unique per BU
REVISION_NUM	NUMBER	Y	0 on first approval; +1 per applied change order
BUSINESS_UNIT_ID	NUMBER	Y	
PO_TYPE	VARCHAR2(15)	Y	STANDARD (Phase 1), RELEASE (Phase 2)
ORIGIN	VARCHAR2(15)	Y	REQUISITION , MANUAL , IMPORT
SUPPLIER_ID / SUPPLIER_SITE_ID	NUMBER	Y	Site must be a purchasing site, not on purchasing hold
SUPPLIER_CONTACT	VARCHAR2(240)	N	
BUYER_ID	NUMBER	Y	
CURRENCY_CODE, RATE_TYPE, RATE_DATE, RATE	-	Y	Rate frozen at approval
PAYMENT_TERMS / PAYMENT_TERMS_ID	VARCHAR2 / NUMBER	Y	Default from supplier site
SHIP_TO_LOCATION_ID / BILL_TO_LOCATION_ID	NUMBER	Y	
DESCRIPTION	VARCHAR2(240)	N	
NOTE_TO_SUPPLIER	VARCHAR2(2000)	N	Printed
DOCUMENT_STATUS	VARCHAR2(20)	Y	6.2.1
CLOSURE_STATUS	VARCHAR2(25)	Y	Rolled up from schedules (6.8)
HOLD_FLAG / HOLD_REASON	VARCHAR2	N	Buyer hold: no receiving or matching while Y
TOTAL_AMOUNT / TOTAL_AMOUNT_FUNC	NUMBER	Y	Excluding tax; maintained by the package
TOTAL_TAX_ESTIMATE	NUMBER	N	Informational (tax is recognised on the invoice)
APPROVAL_REQUEST_ID	NUMBER	N	Current approval cycle
SUBMITTED_DATE / APPROVED_DATE	DATE	N	
COMMUNICATED_DATE	DATE	N	Last time sent to the supplier
ACCEPTANCE_REQUIRED / ACCEPTED_DATE	VARCHAR2(1) / DATE	N	Optional supplier acknowledgement, recorded by the buyer
AGREEMENT_ID	NUMBER	N	Phase 2
AFTER_FACT_FLAG	VARCHAR2(1)	Y	PO created after receipt / invoice date
LEGACY_PO_NUMBER	VARCHAR2(60)	N	Fusion PO number for migrated POs
Audit columns	-	Y	-

RR_PO_LINES

Column	Type	Req	Description
PO_LINE_ID	NUMBER	Y	PK
PO_HEADER_ID	NUMBER	Y	FK
LINE_NUM	NUMBER	Y	
LINE_TYPE	VARCHAR2(10)	Y	QUANTITY / AMOUNT (INVENTORY Phase 2b)
EXPENSE_ITEM_ID	NUMBER	N	
CATEGORY_ID	NUMBER	Y	
ITEM_DESCRIPTION	VARCHAR2(240)	Y	
SUPPLIER_ITEM_NUM	VARCHAR2(60)	N	
UOM_CODE	VARCHAR2(10)	Q	Quantity lines
QUANTITY	NUMBER	Q	Sum of schedule quantities
UNIT_PRICE	NUMBER	Q	Agreed price (quantity lines)
AMOUNT	NUMBER	Y	Quantity × price, or the amount for service lines
TAX_CODE	VARCHAR2(50)	N	Default for schedules
LINE_STATUS	VARCHAR2(15)	Y	OPEN , CLOSED , CANCELLED (roll-up)
CANCEL_REASON	VARCHAR2(400)	N	
CAPEX_FLAG	VARCHAR2(1)	Y	From category
AGREEMENT_LINE_ID	NUMBER	N	Phase 2
NOTE_TO_SUPPLIER	VARCHAR2(1000)	N	
Audit columns	-	Y	-

RR_PO_SCHEDULES — delivery and matching control

Column	Type	Req	Description
SCHEDULE_ID	NUMBER	Y	PK
PO_LINE_ID / PO_HEADER_ID	NUMBER	Y	FK (header denormalised for queries)
SCHEDULE_NUM	NUMBER	Y	
SHIP_TO_LOCATION_ID	NUMBER	Y	
NEED_BY_DATE / PROMISED_DATE	DATE	Y / N	
QUANTITY / AMOUNT	NUMBER	Y	Ordered on this schedule
QUANTITY_RECEIVED / AMOUNT_RECEIVED	NUMBER	Y	Net of returns and corrections
QUANTITY_BILLED / AMOUNT_BILLED	NUMBER	Y	Net of reversed matches
QUANTITY_CANCELLED / AMOUNT_CANCELLED	NUMBER	Y	
MATCH_LEVEL	VARCHAR2(10)	Y	Frozen at approval
ACCRUE_AT_RECEIPT_FLAG	VARCHAR2(1)	Y	Frozen at approval
OVER_RECEIPT_TOL_PCT, INV_QTY_TOL_PCT, INV_PRICE_TOL_PCT, RCV_CLOSE_TOL_PCT, INV_CLOSE_TOL_PCT	NUMBER(5,2)	Y	Frozen at approval (site option → BU option)
TAX_CODE	VARCHAR2(50)	N	

Column	Type	Req	Description
CLOSURE_STATUS	VARCHAR2(25)	Y	OPEN , CLOSED_FOR_RECEIVING , CLOSED_FOR_INVOICING , CLOSED , FINALLY_CLOSED
CANCELLED_FLAG	VARCHAR2(1)	Y	
Audit columns	-	Y	-

RR_PO_DISTRIBUTIONS — the quantity ledger and the accounts

Column	Type	Req	Description
DISTRIBUTION_ID	NUMBER	Y	PK
SCHEDULE_ID / PO_LINE_ID / PO_HEADER_ID	NUMBER	Y	FK (denormalised)
DIST_NUM	NUMBER	Y	
QUANTITY_ORDERED / AMOUNT_ORDERED	NUMBER	Y	
QUANTITY_DELIVERED / AMOUNT_DELIVERED	NUMBER	Y	From receipts
QUANTITY_BILLED / AMOUNT_BILLED	NUMBER	Y	From matches
QUANTITY_CANCELLED / AMOUNT_CANCELLED	NUMBER	Y	
CHARGE_ACCOUNT	VARCHAR2(200)	Y	Expense account
ACCRUAL_ACCOUNT	VARCHAR2(200)	Y	GRNI, frozen at approval from BU options
VARIANCE_ACCOUNT	VARCHAR2(200)	Y	IPV account, frozen (= charge account when the BU option is NULL)
RATE	NUMBER	Y	Frozen PO rate (copied from header at approval)
BUDGET_DATE	DATE	Y	
REQ_DISTRIBUTION_ID	NUMBER	N	Back-link to the requisition
REQUESTER_USER / DELIVER_TO_LOCATION_ID	-	Y	Receiving notifications go to the requester
ACCRUED_AMOUNT_FUNC	NUMBER	Y	Receipt accruals posted, net of matches and write-offs (GRNI balance of this distribution)
ENCUMBERED_AMOUNT	NUMBER	N	Phase 3
Audit columns	-	Y	-

The quantity ledger invariants (enforced by the packages, asserted by the simulation in Appendix A):

```

Σ distributions.QUANTITY_ORDERED      = schedule.QUANTITY      (same for amount)
Σ schedules.QUANTITY                  = line.QUANTITY
QUANTITY_DELIVERED ≤ (ORDERED - CANCELLED) × (1 + OVER_RECEIPT_TOL)    if OVER_RECEIPT_ACTION = REJECT
QUANTITY_BILLED   ≤ basis × (1 + INV_QTY_TOL)    basis = DELIVERED (3-way) or ORDERED - CANCELLED (2-way)
QUANTITY_CANCELLED ≤ ORDERED - max(DELIVERED, BILLED)                  cannot cancel what was received/b.
ACCRUED_AMOUNT_FUNC = Σ receipt accruals - Σ accrual relieved by matches - Σ write-offs (receipt-accrual P
open commitment    = (ORDERED - CANCELLED - BILLED) × price × rate      feeds the commitments report

```

5.3 Revisions and change orders

RR_PO_REVISIONS — one row per approved revision, insert-only.

Column	Type	Req	Description
REVISION_ID	NUMBER	Y	PK
PO_HEADER_ID	NUMBER	Y	
REVISION_NUM	NUMBER	Y	Unique with PO_HEADER_ID
SNAPSHOT_JSON	CLOB	Y	Header, lines, schedules and distributions as approved (IS JSON check)
CHANGE_ORDER_ID	NUMBER	N	NULL for revision 0
CHANGE_SUMMARY	VARCHAR2(2000)	N	Human-readable diff ("Line 2 qty 10 → 12")
Audit columns	-	Y	Insert only

RR_PO_CHANGE_ORDERS — a proposed change to an approved PO. The PO stays open at its current approved values until the change order is approved and applied, so receiving and matching never see half-approved data.

Column	Type	Req	Description
CHANGE_ORDER_ID	NUMBER	Y	PK
PO_HEADER_ID	NUMBER	Y	
CO_NUMBER	VARCHAR2(40)	Y	e.g. PO-DIFC-2026-00017-C01
FROM_REVISION	NUMBER	Y	Revision the change was drafted against (stale check)
STATUS	VARCHAR2(20)	Y	DRAFT , PENDING_APPROVAL , APPLIED , REJECTED , CANCELLED
REASON	VARCHAR2(1000)	Y	
CHANGES_JSON	CLOB	Y	List of changes: {"op": "UPDATE_QTY", "scheduleId": ..., "from": 10, "to": 12}, ADD_LINE , CANCEL_LINE , UPDATE_PRICE , UPDATE_DATE , UPDATE_ACCOUNT , UPDATE_TEXT
AMOUNT_DELTA_FUNC	NUMBER	Y	Effect on the PO total (drives re-approval)
APPROVAL_REQUEST_ID	NUMBER	N	
APPLIED_DATE	DATE	N	
Audit columns	-	Y	-

5.4 Receiving

RR_PO_RCV_HEADERS — one receipt = one delivery note from one supplier (may cover several POs of that supplier).

Column	Type	Req	Description
RECEIPT_HEADER_ID	NUMBER	Y	PK
RECEIPT_NUMBER	VARCHAR2(40)	Y	Unique per BU
BUSINESS_UNIT_ID	NUMBER	Y	
SUPPLIER_ID / SUPPLIER_SITE_ID	NUMBER	Y	
RECEIPT_DATE	DATE	Y	≤ today; open GL period
DELIVERY_NOTE_NUM	VARCHAR2(60)	N	Supplier's packing slip / service sheet number
COMMENTS	VARCHAR2(1000)	N	
RECEIVED_BY	VARCHAR2(150)	Y	
Audit columns	-	Y	-

RR_PO_RCV_TRANSACTIONS — insert-only ledger of receiving events (expense receipts use one-step *receive and deliver*; there is no inspection or put-away for expense lines).

Column	Type	Req	Description
RCV_TRANSACTION_ID	NUMBER	Y	PK
RECEIPT_HEADER_ID	NUMBER	Y	FK
TRANSACTION_TYPE	VARCHAR2(20)	Y	RECEIVE , RETURN_TO_SUPPLIER , CORRECT
PARENT_TRANSACTION_ID	NUMBER	N	The RECEIVE a return or correction refers to
TRANSACTION_DATE	DATE	Y	
PO_HEADER_ID / PO_LINE_ID / SCHEDULE_ID	NUMBER	Y	
QUANTITY / UOM_CODE	NUMBER / VARCHAR2	Q	Signed: returns and negative corrections are negative
AMOUNT	NUMBER	Y	Quantity × PO price, or the received amount (service lines); signed
CURRENCY_CODE / RATE	-	Y	PO currency, PO rate (accrual is at PO rate)
AMOUNT_FUNC	NUMBER	Y	
REASON_CODE	VARCHAR2(30)	N	Returns: DAMAGED , WRONG_ITEM , NOT_ORDERED , QUALITY , OTHER
ACCOUNTING_STATUS	VARCHAR2(15)	Y	UNACCOUNTED , ACCOUNTED , NOT_REQUIRED (period-end-accrual POs)
SLA_HEADER_ID / GL_BATCH_ID	NUMBER	N	Written back by Create Accounting
Audit columns	-	Y	Insert only

RR_PO_RCV_DISTRIBUTIONS — how each receiving event is spread over the schedule's distributions (pro-rata to open quantity; the last distribution absorbs rounding). This is what the accrual and the 3-way match use.

Column	Type	Req	Description
RCV_DIST_ID	NUMBER	Y	PK
RCV_TRANSACTION_ID	NUMBER	Y	FK
DISTRIBUTION_ID	NUMBER	Y	FK
QUANTITY / AMOUNT / AMOUNT_FUNC	NUMBER	Y	Signed
CHARGE_ACCOUNT / ACCRUAL_ACCOUNT	VARCHAR2(200)	Y	Copied from the distribution
QUANTITY_BILLED / AMOUNT_BILLED	NUMBER	Y	Matched against this receipt (3-way consumption)
Audit columns	-	Y	-

5.5 Invoice matching (AP integration)

RR_PO_INVOICE_MATCHES — every link between an AP invoice line and a PO distribution (and, for 3-way, the receipt distribution consumed). Insert-only; a match is undone by a reversing row.

Column	Type	Req	Description
MATCH_ID	NUMBER	Y	PK
INVOICE_ID / INVOICE_LINE_NUMBER	NUMBER	Y	RR_AP_INVOICE_LINES_ALL
PO_HEADER_ID / PO_LINE_ID / SCHEDULE_ID / DISTRIBUTION_ID	NUMBER	Y	

Column	Type	Req	Description
RCV_DIST_ID	NUMBER	N	Receipt consumed (<code>THREE_WAY</code>); NULL while matched ahead of receipt (held <code>QTY_REC</code>), filled when received
MATCH_TYPE	VARCHAR2(10)	Y	<code>PO</code> (2-way) / <code>RECEIPT</code> (3-way)
QUANTITY / AMOUNT	NUMBER	Y	Invoiced, in invoice currency; signed (credit memos negative)
INVOICE_UNIT_PRICE / PO_UNIT_PRICE	NUMBER	Q	
INVOICE_RATE / BASE_RATE	NUMBER	Y	Base = PO rate (2-way) or receipt rate (3-way)
DEBIT_ACCOUNT	VARCHAR2(200)	Y	<code>ACCRUAL_ACCOUNT</code> if accrued at receipt, else <code>CHARGE_ACCOUNT</code>
BASE_AMOUNT_FUNC	NUMBER	Y	Quantity × PO price × base rate (relieves the accrual)
PRICE_VARIANCE_FUNC	NUMBER	Y	(invoice price – PO price) × qty × invoice rate
EXCHANGE_VARIANCE_FUNC	NUMBER	Y	Quantity × PO price × (invoice rate – base rate)
REVERSAL_OF_MATCH_ID	NUMBER	N	Set on the reversing row
MATCH_STATUS	VARCHAR2(10)	Y	<code>ACTIVE</code> , <code>REVERSED</code>
Audit columns	-	Y	Insert only (status updated once on reversal)

RR_AP_INVOICE_HOLDS (Payables-owned, created by Procurement script 3xx) — an unreleased hold stops the invoice from being accounted and paid.

Column	Type	Req	Description
HOLD_ID	NUMBER	Y	PK
INVOICE_ID	NUMBER	Y	
INVOICE_LINE_NUMBER	NUMBER	N	NULL = header-level hold
HOLD_CODE	VARCHAR2(30)	Y	6.6.3
HOLD_REASON	VARCHAR2(1000)	Y	Generated text, e.g. "Billed 12 > received 10 (tol 0%)"
HELD_BY	VARCHAR2(150)	Y	<code>SYSTEM</code> for validation holds
HOLD_DATE	DATE	Y	
RELEASE_CODE	VARCHAR2(30)	N	<code>RESOLVED</code> (system, condition cleared), <code>APPROVED_VARIANCE</code> , <code>QTY_OK</code> , <code>PRICE_OK</code> , <code>OTHER</code>
RELEASE_REASON	VARCHAR2(1000)	N	
RELEASED_BY / RELEASE_DATE	-	N	
Audit columns	-	Y	-

5.6 Accruals

RR_PO_ACCRUAL_RUNS — period-end accrual for POs with `ACCRUE_AT_RECEIPT_FLAG` = N .

Column	Type	Req	Description
RUN_ID	NUMBER	Y	PK
BUSINESS_UNIT_ID / PERIOD_NAME	-	Y	One <code>POSTED</code> run per BU and period
ACCRUAL_DATE	DATE	Y	Last day of the period
REVERSAL_DATE	DATE	Y	First day of the next period

Column	Type	Req	Description
STATUS	VARCHAR2(15)	Y	DRAFT (review), POSTED , CANCELLED
TOTAL_AMOUNT_FUNC	NUMBER	Y	
SLA_HEADER_ID / GL_BATCH_ID / REVERSAL_GL_BATCH_ID	NUMBER	N	
Audit columns	-	Y	-

RR_PO_ACCRUAL_LINES — one per distribution with received-not-billed value at period end.

Column	Type	Req	Description
ACCRUAL_LINE_ID	NUMBER	Y	PK
RUN_ID / DISTRIBUTION_ID	NUMBER	Y	
QUANTITY / AMOUNT_FUNC	NUMBER	Y	(delivered – billed) at the period end, valued at PO price × PO rate
CHARGE_ACCOUNT / ACCRUAL_ACCOUNT	VARCHAR2(200)	Y	
Audit columns	-	Y	Insert only

RR_PO_ACCRUAL_WRITE_OFFS — clears receipt accruals that will never be invoiced (supplier waived, PO closed under-billed).

Column	Type	Req	Description
WRITE_OFF_ID	NUMBER	Y	PK
DISTRIBUTION_ID	NUMBER	Y	
AMOUNT_FUNC	NUMBER	Y	≤ the distribution's ACCRUED_AMOUNT_FUNC
WRITE_OFF_ACCOUNT	VARCHAR2(200)	Y	Default BU option, or back to the charge account
WRITE_OFF_DATE / REASON	DATE / VARCHAR2(1000)	Y	
SLA_HEADER_ID / GL_BATCH_ID	NUMBER	N	
Audit columns	-	Y	Insert only

5.7 Cross-cutting and reserved tables

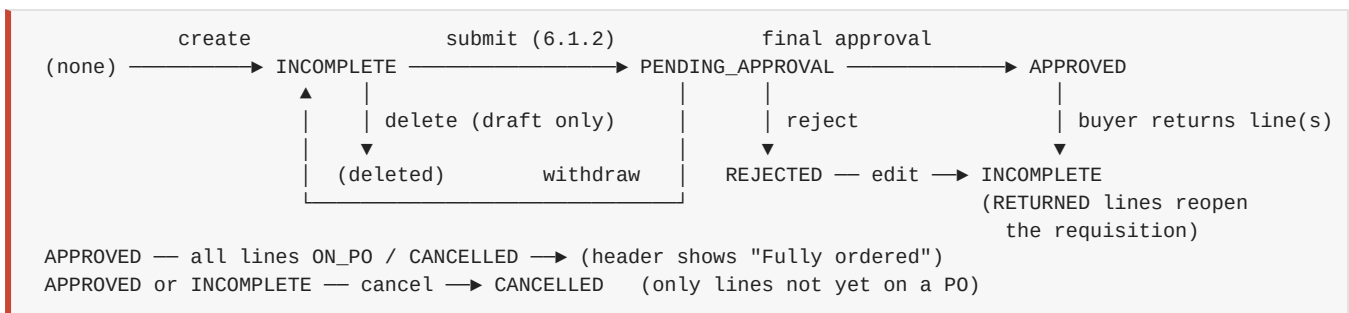
Table	Key columns	Notes
RR_PO_ATTACHMENTS	ATTACHMENT_ID, ENTITY_TYPE (REQ , PO , CO , RCV), ENTITY_ID, FILE_NAME, MIME_TYPE, FILE_BLOB, CATEGORY (QUOTATION , SPECIFICATION , DELIVERY_NOTE , OTHER), SEND_TO_SUPPLIER_FLAG	Same upload/download endpoint pattern as AP payment attachments; files flagged SEND_TO_SUPPLIER_FLAG are attached to the PO email
RR_PO_ACTION_HISTORY	HISTORY_ID, ENTITY_TYPE, ENTITY_ID, ACTION, FROM_STATUS, TO_STATUS, ACTION_BY, ACTION_DATE, COMMENTS	Insert-only audit of every action (submit, approve, return, cancel, close, communicate ...); approvals also keep RR_APPROVAL_HISTORY
RR_PO_COMMUNICATIONS	COMM_ID, PO_HEADER_ID, REVISION_NUM, METHOD, TO_EMAIL, CC_EMAIL, SENT_DATE, STATUS, ERROR_TEXT, PDF_ATTACHMENT_ID	Uses the existing server-side email (Brevo) used by approvals

Table	Key columns	Notes
RR_PO_PROC_REGISTRY	PROC_NAME, PARAMS_CSV, ENABLED_FLAG, REQUIRED_ROLE	Write-gateway whitelist (9.2)
RR_PO_EXEC_LOG	LOG_ID, USER_NAME, PROC_NAME, PARAMS_CLOB, STATUS, MESSAGE, ELAPSED_MS	Every gateway call
RR_PO_AGREEMENTS / _LINES (Phase 2)	Supplier, site, currency, start/end dates, amount limit, released amount; lines: item/category, price, price breaks	POs of type RELEASE reference AGREEMENT_ID / AGREEMENT_LINE_ID ; requisition lines can be sourced from them
RR_PO_BUDGETS / RR_PO_FUNDS_LEDGER (Phase 3)	Budget by BU, period, account mask; funds ledger rows of type PRECOMMITMENT (req), COMMITMENT (PO), OBLIGATION (invoice), ACTUAL , each with a relieving row	Funds available = budget – Σ unrelieved rows

6. Processes and Business Rules

6.1 Requisitions

6.1.1 Requisition status model



6.1.2 Submit — validations and steps

1. Requester and preparer have module access and BU access; the BU has ACTIVE options.
2. At least one line; every line has category, description, need-by date $\geq$ today, deliver-to location, amount > 0 .
3. Quantity lines have UOM, quantity > 0 , unit price ≥ 0 ; amount lines have amount > 0 .
4. Every line has distributions summing to 100 %; every charge account is a valid, enabled combination whose segment 1 equals the BU's company (RR_GL_BUSINESS_UNITS.COMPANY).
5. Rate available for every foreign-currency line on the rate date (RR_CURRENCY_DAILY_RATES); compute AMOUNT_FUNC , header TOTAL_AMOUNT_FUNC .
6. Assign REQ_NUMBER (4.9) if not yet assigned.
7. Phase 3: funds check per distribution when BUDGET_CONTROL_LEVEL $\neq$ NONE .
8. If REQ_APPROVAL_REQUIRED = N → APPROVED directly; else create the approval request (6.10) → PENDING_APPROVAL .
9. Write RR_PO_ACTION_HISTORY .

6.1.3 Charge-account derivation

```

start    = RR_PO_REQUESTER_DEFAULTS.CHARGE_ACCOUNT_TEMPLATE (requester + BU)
seg 1    = RR_GL_BUSINESS_UNITS.COMPANY of the requisition BU
seg 4    = EXPENSE_ITEM.NATURAL_ACCOUNT_OVERRIDE
           else CATEGORY.DEFAULT_NATURAL_ACCOUNT (walk up the category tree)
           else keep the template's segment 4 (warning: "category has no account")
then     = apply RR_PO_ACCOUNT_RULES matching (BU, category or ancestor, item), by PRIORITY, per segment
finally  = validate the combination exists and is enabled; if the combination does not exist
           but every segment value is valid, create it through the existing GL combination
           procedure (same behaviour as AP distribution entry), else error REQ-08

```

The derived account is shown on the line; users with `EDIT_ACCOUNT` can change it, and the change is kept.

6.1.4 Buyer assignment (on approval)

First match wins: buyer specialised in the line's category (or an ancestor) for the BU → buyer for the BU with no category → BU default buyer (`DEFAULT_FLAG = Y`). No buyer found → line stays unassigned and appears in the "Unassigned requisition lines" queue of the procurement manager.

6.2 Purchase orders

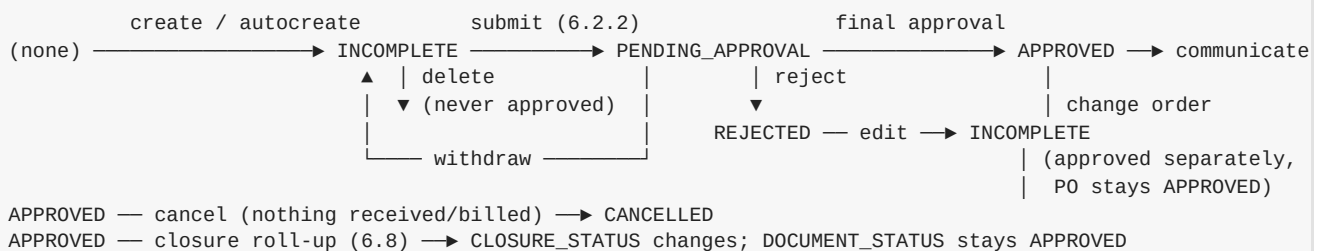
6.2.0 Creating a PO directly (no requisition)

This is the standard way to buy. **Create Purchase Order** (`/po/orders/new`):

1. **Header** — BU (from the user's BU access), supplier and site (LOVs from the existing `RR_SUPPLIER_*` tables, filtered to sites usable by the BU), currency (site default, else functional), payment terms (site), ship-to / bill-to (site assignment, else BU defaults), buyer (the user, when they are a buyer), description.
2. **Lines** — pick a catalog item or type free text; category; quantity × price or amount; need-by date; tax code (category / item default). One schedule per line is created automatically (more can be added for split deliveries).
3. **Charge account** — derived exactly as for requisitions (6.1.3) using the line's **requester**, which defaults to the person creating the PO (or can be set to the employee the purchase is for); editable with `EDIT_ACCOUNT`; split into several distributions by percentage when the cost is shared.
4. **Save** (`INCOMPLETE`, PO number assigned) → **Submit** → approval (6.10) → approved → sent to the supplier.

From here a direct PO is identical to one created from a requisition: same receiving, matching, accruals, change orders and close. Its distributions simply have no `REQ_DISTRIBUTION_ID` and the header has `ORIGIN = MANUAL`. Copy-PO (from any earlier PO, with new dates) makes repeat purchases one click.

6.2.1 PO document status model



Receiving and matching are allowed only when `DOCUMENT_STATUS = APPROVED`, `HOLD_FLAG = N` and the schedule's closure status allows it (receiving: `OPEN` or `CLOSED_FOR_INVOICING`; invoicing: `OPEN` or `CLOSED_FOR_RECEIVING`).

6.2.2 Submit PO — validations and steps

1. Buyer is an active buyer for the BU; supplier and site valid for the BU per 4.10 (existing `RR_SUPPLIER_*` tables), not on purchasing hold.
2. Currency rate available; payment terms set; ship-to and bill-to locations valid for the BU.

3. Lines, schedules and distributions complete; quantity ledger sums consistent (5.2 invariants); charge accounts valid; tax codes active and assigned to the BU (`RR_TAX_ASSIGNMENTS`).
4. Freeze on each schedule: `MATCH_LEVEL` (category `RECEIPT_REQUIRED_FLAG` → site option → BU option by line type), `ACCUE_AT_RECEIPT_FLAG` , all tolerances; on each distribution: `ACCRUAL_ACCOUNT` , `VARIANCE_ACCOUNT` , `RATE` .
5. Separation of duties: the submitter cannot be the only approver (approval engine skips the submitter and escalates to the next approver in the rule).
6. Approval request (6.10) or direct approval when `PO_APPROVAL_REQUIRED` = N (or amount below the lowest rule).
7. On approval: write `RR_PO_REVISIONS` (revision 0), set requisition lines `ON_PO` , notify the buyer, and if the site's `PO_COMMUNICATION` = `EMAIL` send the PO automatically (6.2.3).

6.2.3 Communicate to supplier

The PO PDF is generated client-side with the existing jsPDF stack (same approach as the P&L and payment PDFs): company letterhead, PO number and revision, supplier and site address, ship-to and bill-to, buyer contact, lines with schedules, tax estimate, total, payment terms and `PO_TERMS_TEXT` . The page uploads it as a `RR_PO_ATTACHMENTS` row, then calls `RR_PO_DOC_PKG.COMMUNICATE` which emails it through the existing server-side mail service and logs `RR_PO_COMMUNICATIONS` . Revised POs print "Revision n" and a change summary.

6.3 Autocreate (requisition → PO)

The buyer workbench lists approved, unassigned-to-PO requisition lines for the buyer's BUs. The buyer selects lines, sets or confirms the supplier, site and price, and creates POs:

```
group key = BU + supplier + supplier site + currency + ship-to → one PO per group
lines     = one PO line per requisition line; requisition lines with the same item/description,
           UOM, price and need-by may be combined into one PO line (option on the workbench)
schedules = one per requisition line (need-by, deliver-to)
dists     = copied 1:1 from requisition distributions (REQ_DISTRIBUTION_ID kept)
status    = INCOMPLETE (buyer reviews, then submits) – or submitted immediately in AUTO mode
```

`AUTOCREATE_MODE` = `AUTO` creates and submits POs without a buyer step for approved catalog lines that have a preferred supplier and site and a list price; anything else falls back to the workbench.

6.4 Change orders

Change	Allowed when	Re-approval
Increase quantity / amount	Schedule not finally closed	If total increase > <code>CO_REAPPROVAL_THRESHOLD_PCT</code>
Decrease quantity / amount	New value ≥ max(delivered, billed)	No
Change unit price	Nothing billed on the line	If total increases beyond the threshold
Change need-by / promised date, notes	Always (until closed)	No
Change charge account	Nothing delivered or billed on the distribution	No (logged)
Add a line	Always	Yes (counts as an increase)
Cancel a line / schedule	Nothing delivered or billed (else decrease to delivered)	No
Change supplier or currency	Never — cancel and re-create	-

Applying a change order (on approval, or immediately when no approval is needed): lock the PO (`FOR UPDATE`), check `FROM_REVISION` = `REVISION_NUM` (else "PO changed since this change order was drafted"), apply every change in `CHANGES_JSON` , re-run the PO validations, `REVISION_NUM + 1` , insert `RR_PO_REVISIONS` , re-communicate.

6.5 Receiving

6.5.1 Receive

The receiver searches open schedules (by PO number, supplier, requester or deliver-to) and enters the quantity (quantity lines) or amount (amount lines) received, per schedule, under one receipt header.

1. PO approved, not on hold; schedule `OPEN` or `CLOSED_FOR_INVOICING`; match level irrelevant (2-way schedules can still be received, for example to confirm a service).
2. Receipt date $\leq$ today, not before the PO approval date, in an open GL period; early-receipt check (`EARLY_RECEIPT_DAYS`).
3. Over-receipt: $\text{delivered} + \text{new} \leq (\text{ordered} - \text{cancelled}) \times (1 + \text{tol})$; `REJECT` → error, `WARNING` → allowed with a warning in the response and in the action history.
4. Insert `RR_PO_RCV_TRANSACTIONS` (`RECEIVE`) and `RR_PO_RCV_DISTRIBUTIONS` (pro-rata to each distribution's open quantity); update `QUANTITY_DELIVERED` / `AMOUNT_DELIVERED` on distributions and `QUANTITY_RECEIVED` / `AMOUNT_RECEIVED` on the schedule.
5. Accounting status: `UNACCOUNTED` when `ACCRUE_AT_RECEIPT_FLAG = Y`, else `NOT_REQUIRED`.
6. Closure roll-up (6.8). Notify the requester ("Your order PO-... has been received").

6.5.2 Return to supplier

Against a `RECEIVE` transaction: quantity $\leq$ received – already returned – already billed against that receipt (3-way). Creates negative rows, reduces delivered quantities, reverses the accrual pro-rata (accounting 7.3), reopens the schedule for receiving if it had auto-closed. A reason code is mandatory. Optionally the AP clerk raises a debit memo (existing AP credit-memo flow, matched to the PO with negative quantity).

6.5.3 Correction

Positive or negative adjustment of a `RECEIVE` (typing error). Negative corrections obey the same limit as returns; positive corrections obey the over-receipt rule. Accounting as a receipt or return.

6.6 Invoice matching (in Payables)

6.6.1 Matching from the AP invoice

`Create Invoice` gains a **Match to PO** mode (screen in 10.2). The AP clerk picks the supplier site, then:

- **2-way schedules:** pick PO schedules; enter quantity (or amount) and invoice unit price.
- **3-way schedules:** pick receipts (receipt lines of those schedules with unbilled quantity); enter quantity and price. The match consumes `RR_PO_RCV_DISTRIBUTIONS.QUANTITY_BILLED` FIFO by receipt date unless a specific receipt is chosen. If the invoice arrives before the goods, the not-yet-received quantity may still be matched to the schedule: it is recorded without a receipt link and the invoice is held `QTY_REC`; when the receipt is entered the link is filled and the hold releases automatically.

For each matched quantity the package creates the AP invoice line (line type `Item`, `PURCHASE_ORDER_NUMBER`, `PURCHASE_ORDER_LINE_NUMBER`, `RECEIPT_NUMBER`, `RECEIPT_LINE_NUMBER` filled, `DISTRIBUTION_COMBINATION` = the match's `DEBIT_ACCOUNT`, `TAX_CLASSIFICATION` = schedule tax code), one `RR_PO_INVOICE_MATCHES` row per distribution (pro-rata), and increases billed quantities on distributions, schedules and receipt distributions. Tax lines are calculated by the existing AP tax logic from the tax code.

Freight or miscellaneous charges on the invoice that are not on the PO are entered as normal non-matched AP lines.

6.6.2 Invoice validation (holds)

`RR_PO_MATCH_PKG.VALIDATE_INVOICE(p_invoice_id)` runs when the invoice is saved, re-validated, or a related PO, receipt or match changes. It **releases** system holds whose condition has cleared (`RELEASE_CODE = RESOLVED`) and **places** holds for every failing condition:

6.6.3 Hold codes

Hold code	Condition (per schedule, all matched invoices together)	Typical resolution
QTY_ORD	billed > (ordered – cancelled) × (1 + inv qty tol)	Change order raising the quantity, or credit memo
QTY_REC	3-way: billed > received × (1 + inv qty tol)	Receive the goods; hold auto-releases
PRICE	invoice price > PO price × (1 + inv price tol)	Change order on price, or manual release APPROVED_VARIANCE
AMOUNT	amount lines: billed amount > ordered amount × (1 + tol) or > INVOICE_AMOUNT_TOLERANCE	Change order or release
PO_NOT_APPROVED	PO on hold, or a pending change order affects the matched schedule	Approve the change order / remove hold
FINAL_MATCH	schedule finally closed after the invoice was matched	Reverse the match
CURRENCY	invoice currency ≠ PO currency	Re-enter the invoice in PO currency

Manual holds (MANUAL) can also be placed by AP. **An invoice with any unreleased hold cannot be accounted, cannot be selected for payment, and is shown with a red "On hold" tag** in Manage Invoices. The payment selection (rr_ap_available_installments_get) and Create Accounting for AP invoices add a NOT EXISTS on unreleased holds.

6.6.4 Variances

For every match row (functional currency, rounded at the row):

```

base_func      = qty × PO_price × base_rate          base_rate = PO rate (2-way) / receipt rate (3-way)
IPV            = qty × (invoice_price – PO_price) × invoice_rate
ERV           = qty × PO_price × (invoice_rate – base_rate)
invoice_func   = qty × invoice_price × invoice_rate = base_func + IPV + ERV          (identity)

```

IPV posts to the distribution's VARIANCE_ACCOUNT (the charge account unless the BU sets a central IPV account); ERV posts to the BU exchange gain or loss account. For period-end-accrual POs ERV is still separated so expense stays at the PO value; this is configurable (open decision D6).

6.7 Accruals

6.7.1 Receipt accrual (ACCRUE_AT_RECEIPT_FLAG = Y)

The receipt books the cost and the liability immediately (Dr expense, Cr GRNI). The invoice match relieves GRNI. The **uninvoiced receipts** report (= GRNI sub-ledger) is Σ ACCRUED_AMOUNT_FUNC per distribution and must equal the GL balance of the accrual account — the reconciliation report shows both and the difference.

6.7.2 Period-end accrual (ACCRUE_AT_RECEIPT_FLAG = N)

Receipts create no accounting. At period end, **Run Period-End Accrual** for a BU and period:

```

for each distribution with ACCRUE_AT_RECEIPT_FLAG = N:
  qty_unbilled = delivered_as_of(period_end) – billed_as_of(period_end)      (as-of dates, not today's count)
  if qty_unbilled > 0: accrual line = qty_unbilled × PO price × PO rate
DRAFT run → review grid (by supplier / PO / account) → Post:
  journal on ACCRUAL_DATE  Dr charge account / Cr accrual account
  reversal on REVERSAL_DATE Cr charge account / Dr accrual account  (auto-reversing)

```

One posted run per BU and period; re-running cancels the previous draft. A posted run can be cancelled only before the period closes (creates reversing entries).

6.7.3 Accrual write-off

For receipt-accrual distributions that are closed (or older than `ACCRUAL_WRITE_OFF_AGE_DAYS`) with `ACCRUED_AMOUNT_FUNC > 0`: select, enter reason, post Dr GRNI / Cr write-off account. The distribution's accrued balance is reduced; the PO can then be finally closed.

6.8 Close and cancel

Closure is evaluated per schedule after every receipt, return, correction, match, match reversal and cancel:

```
rcv_done = received ≥ (ordered - cancelled) × (1 - RCV_CLOSE_TOL)    or match level is TWO_WAY and billed
inv_done = billed ≥ basis × (1 - INV_CLOSE_TOL)                    basis = received (3-way) / ordered

rcv_done and inv_done → CLOSED
rcv_done only         → CLOSED_FOR_RECEIVING
inv_done only         → CLOSED_FOR_INVOICING
neither               → OPEN
FINALLY_CLOSED is never set automatically and never re-opened
```

Automatic statuses are recomputed on every event, so they move both ways: a 3-way schedule with 12 received and 12 billed (of 20) is `CLOSED_FOR_INVOICING`, and becomes `OPEN` again when 3 more are received (simulation A2).

Line and header closure statuses are the "least closed" of their children (`OPEN < CLOSED_FOR_RECEIVING / CLOSED_FOR_INVOICING < CLOSED < FINALLY_CLOSED`, cancelled children ignored). Manual actions:

- **Close / reopen** (buyer) — sets or clears a manual close on a schedule, line or header.
- **Finally close** (procurement manager) — irreversible; requires no unreleased holds on matched invoices and, for receipt-accrual distributions, `ACCRUED_AMOUNT_FUNC = 0` (write off first). Releases the remaining commitment (and Phase 3 encumbrance).
- **Cancel** (buyer) — header, line or schedule; only the unreceived and unbilled part is cancelled (`QUANTITY_CANCELLED = ordered - max(delivered, billed)`); a requisition line linked to a cancelled PO line can be returned to the buyer pool (`RECREATE_DEMAND = Y`) or cancelled.

6.9 Budgetary control hooks (Phase 3)

Every requisition and PO distribution carries `BUDGET_DATE`; packages call `RR_PO_BUDGET_PKG.CHECK_FUNDS / RESERVE_FUNDS` at submit and approval when the BU's `BUDGET_CONTROL_LEVEL ≠ NONE`. Phase 1 implements these as no-ops returning `PASSED`, so enabling Phase 3 needs no change to the requisition or PO code paths.

6.10 Approvals (existing engine, extended)

Requisitions, POs and change orders use the existing approval engine:

Document	MODULE	TRANSACTION_TYPE	TRANSACTION_ID	AMOUNT
Requisition	PROCUREMENT	REQUISITION	REQ_HEADER_ID	TOTAL_AMOUNT_FUNC
Purchase order	PROCUREMENT	PURCHASE_ORDER	PO_HEADER_ID	TOTAL_AMOUNT_FUNC
Change order	PROCUREMENT	PO_CHANGE_ORDER	CHANGE_ORDER_ID	new PO total (func)

Two additive extensions to the engine (script 3xx; existing AP/AR/Cash rules keep working unchanged):

1. `RR_APPROVAL_RULES` gets nullable `BUSINESS_UNIT_ID` and `CATEGORY_CODE`. Rule selection adds `(rule.BUSINESS_UNIT_ID IS NULL OR = doc BU)` and, for requisitions/POs, `(rule.CATEGORY_CODE IS NULL OR the document has a line in that category or a child of it)`; still ordered by `PRIORITY`.
2. `RR_APPROVAL_CALLBACKS` (`MODULE, TRANSACTION_TYPE, CALLBACK_PROC`): when a request reaches a final state (`APPROVED, REJECTED, RECALLED, CANCELLED`), the engine calls `CALLBACK_PROC(p_request_id,`

`p_transaction_id, p_decision, p_user`). Procurement registers `RR_PO_APPROVAL_PKG.ON_DECISION`, which moves the document to its next status. Email links, tokens and the Requests tab work as today.

No rule found → error "No approval rule for PROCUREMENT/REQUISITION, BU ..., amount ..." unless the BU option says approval is not required.

7. Accounting

7.1 Choosing the accrual method

	Receipt accrual (Y)	Period-end accrual (N)
When cost hits the P&L	At receipt	At invoice, plus month-end accrual journal (auto-reversed)
Uninvoiced liability visible	Always (GRNI account)	Only at month end
Entries per order	Receipt + match	Invoice only (+ monthly accrual)
Best for	Goods and services where receipt date matters, strict cut-off	High-volume low-value expense, services billed monthly
Re-ERP default	Y for quantity lines , configurable (open decision D1)	

7.2 Event types

Event	Source	SLA SOURCE_TABLE / EVENT_TYPE_CODE	GL REFERENCE ⁵
Receive	<code>RR_PO_RCV_TRANSACTIONS</code>	<code>RR_PO_RCV_TRANSACTIONS / PO_RECEIVE</code>	<code>PO_RECEIPTS</code>
Return	same	<code>PO_RETURN</code>	<code>PO_RECEIPTS</code>
Correction	same	<code>PO_CORRECT</code>	<code>PO_RECEIPTS</code>
Period-end accrual	<code>RR_PO_ACCRUAL_RUNS</code>	<code>RR_PO_ACCRUAL_RUNS / PO_PERIOD_END_ACCRUAL</code> (+ reversal)	<code>PO_ACCRUALS</code>
Accrual write-off	<code>RR_PO_ACCRUAL_WRITE_OFFS</code>	<code>PO_ACCRUAL_WRITE_OFF</code>	<code>PO_ACCRUALS</code>
Matched invoice	existing AP invoice event	<code>AP_INVOICES</code> (existing) — distribution accounts come from the match	<code>AP_INVOICES</code>

REFERENCE² = the source id (`RCV_TRANSACTION_ID`, `RUN_ID`, `WRITE_OFF_ID`), the convention AP, AR, Cash and Inventory use, so Check Accounting and journal drill-down link back automatically. Create Accounting for receipts groups `UNACCOUNTED` receiving transactions by BU and accounting date into one batch, writes `SLA_HEADER_ID / GL_BATCH_ID` back and sets `ACCOUNTED`. Unbalanced batches are impossible by construction and are also rejected by the existing GL journal guard.

7.3 Accounting rules

Event	Debit	Credit	Amount (functional)
Receive (accrue at receipt)	Charge account	Accrual (GRNI)	qty × PO price × PO rate
Return / negative correction	Accrual (GRNI)	Charge account	same, negative event reversed

Event	Debit	Credit	Amount (functional)
Invoice match, accrued PO	Accrual (GRNI) — base amount	AP liability (existing) — invoice total	base_func
— price variance	Variance account (IPV)		IPV (credit if negative)
— exchange variance	Exchange loss	(or Cr exchange gain)	ERV
— recoverable input tax	Tax account (RR_TAX_ASSIGNMENTS.TAX_ACCOUNT)		tax amount (existing AP logic)
Invoice match, period-end PO	Charge account — base	AP liability	base_func
— price variance	Variance account (IPV; = charge account unless a central IPV account is set)		IPV
— exchange variance	Exchange loss / gain		ERV
Period-end accrual	Charge account	Accrual account	(delivered – billed) × PO price × PO rate
— automatic reversal (next period)	Accrual account	Charge account	same
Accrual write-off	Accrual (GRNI)	Write-off account	written-off amount
PO approval / cancel	—	—	no accounting in Phase 1 (encumbrance in Phase 3)

7.4 Worked examples (BU DIFC, functional currency AED, VAT 5 % recoverable)

A. Office chairs — quantity line, 3-way, receipt accrual

PO: 20 chairs × 450.00 AED = 9,000.00, charge 01-10-200-6105001-... (Office furniture expense).

Step	Entry
Receive 20	Dr Office furniture expense 9,000.00 / Cr GRNI 9,000.00
Invoice: 20 × 460.00 + VAT 5 % = 9,660.00 (price tol 5 %: +2.2 % OK)	Dr GRNI 9,000.00 · Dr Office furniture expense (IPV) 200.00 · Dr Input VAT 460.00 / Cr AP liability 9,660.00
Result	GRNI 0.00; expense 9,200.00 (actual price); PO CLOSED

B. Annual AC maintenance — amount line, 2-way, four quarterly schedules of 3,000.00

Step	Entry
Q1 invoice 3,000.00 + VAT 150.00, matched to schedule 1	Dr Repairs & maintenance 3,000.00 · Dr Input VAT 150.00 / Cr AP liability 3,150.00
Schedule 1	billed 3,000 = ordered → CLOSED (2-way: receiving considered done by billing)

C. Software licences — USD PO, receipt accrual

PO 10 × 120.00 USD, PO rate 3.6725. Receipt at PO rate. Invoice 10 × 120.00 at rate 3.6800.

Step	Entry
Receive 10	Dr Software expense 4,407.00 / Cr GRNI 4,407.00 (1,200 × 3.6725)

Step	Entry
Invoice 1,200.00 USD = 4,416.00 AED (+ VAT)	Dr GRNI 4,407.00 · Dr Exchange loss 9.00 · Dr Input VAT 220.80 / Cr AP liability 4,636.80

D. Return

Of example A, 2 chairs returned before invoicing: Dr GRNI 900.00 / Cr Office furniture expense 900.00. Invoice for 18 chairs matches the remaining 18 received.

E. Period-end accrual

Stationery PO (accrue at receipt N), 500.00 received 28 Sep, invoice arrives 5 Oct.

Date	Entry
30 Sep (accrual run)	Dr Stationery expense 500.00 / Cr Accrued liabilities 500.00
1 Oct (auto-reversal)	Dr Accrued liabilities 500.00 / Cr Stationery expense 500.00
5 Oct invoice	Dr Stationery expense 500.00 · Dr Input VAT 25.00 / Cr AP liability 525.00

All five examples are executed by the simulation in Appendix A, which checks that every journal balances, GRNI returns to zero, and the identity $\text{invoice} = \text{base} + \text{IPV} + \text{ERV}$ holds.

7.5 Reconciliation

- **GRNI reconciliation:** $\text{RR_PO_V_ACCRUAL_RECON} = \sum \text{ACCRUED_AMOUNT_FUNC}$ by accrual account and BU, versus the GL balance of the same account (from the TB service) → difference with drill to the receipts/matches that are unaccounted or posted outside Procurement.
- **Commitments:** open commitment per PO distribution (5.2) by supplier, category, cost centre.
- **Three-way status:** per schedule ordered / received / billed / cancelled / open with holds.

8. Security and Access

Privilege (module PO)	Requester	Buyer	Receiver	Proc. manager	AP clerk	Admin
Create / submit own requisitions	✓	✓		✓		✓
Create purchase orders directly (6.2.0)		✓		✓		✓
Requisition on behalf of others		✓		✓		✓
EDIT_ACCOUNT on lines		✓		✓		✓
Process requisitions / create POs		✓		✓		✓
Change orders, cancel, close		✓		✓		✓
Finally close, write off accruals				✓		✓
Receive / return / correct	own lines	✓	✓	✓		✓
Match invoices, release holds					✓	✓
Period-end accrual, reconciliation				✓	✓	✓
Setup				✓		✓

- Module access and BU scoping reuse `RR_USER_MODULE_ACCESS` and `RR_USER_BU_ACCESS`: every package checks the user's BU access for the document's BU, every read view is filtered by the user's BUs.
- Requesters see their own requisitions and receipts; buyers see documents of their BUs; approvers see what they approve through the approval engine.

- Segregation of duties enforced in packages: an approver cannot approve a document they submitted; a receiver cannot receive on a PO they are the buyer of when the BU option `SOD_BUYER_RECEIVE = N`; the user who matched an invoice cannot release its price hold.

9. Service Architecture

Identical to the Inventory RD section 13: no per-feature ORDS endpoints. Two gateways, all logic in packages.

9.1 Read path — `POST ai/executequery` (existing), `appUser` `PROCUREMENT`

View	Feeds
<code>RR_PO_V_REQUISITIONS</code> / <code>RR_PO_V_REQ_LINES</code>	My Requisitions, requisition detail, buyer workbench
<code>RR_PO_V_ORDERS</code>	Purchase Orders grid (header + supplier + buyer + totals + closure)
<code>RR_PO_V_ORDER_LINES</code>	PO detail: lines → schedules → distributions with the quantity ledger
<code>RR_PO_V_OPEN_SCHEDULES</code>	Receive screen, invoice matching (2-way)
<code>RR_PO_V_RECEIPTS</code>	Receipts grid, return/correct, invoice matching (3-way, with unbilled qty)
<code>RR_PO_V_MATCHES</code>	Invoice ↔ PO drill (both directions)
<code>RR_PO_V_INVOICE_HOLDS</code>	Invoice holds workbench
<code>RR_PO_V_CHANGE_ORDERS</code> / <code>RR_PO_V_REVISIONS</code>	Change history
<code>RR_PO_V_ACCRUALS</code> / <code>RR_PO_V_ACCRUAL_RECON</code>	Uninvoiced receipts, GRNI reconciliation
<code>RR_PO_V_SPEND</code>	Spend analysis (matched invoices + receipts by category/supplier/cost centre/month)
<code>RR_PO_V_SETUP_*</code>	Setup pages (one per setup table, with resolved names)

9.2 Write path — `POST po/execute` (one new endpoint)

Same handler template as `inv/execute`: look the procedure up in `RR_PO_PROC_REGISTRY`, bind every JSON parameter by name (never concatenated), execute, log to `RR_PO_EXEC_LOG`, return `{ "success": true, "id": ..., "number": "PO-DIFC-2026-00017", "message": "...", "warnings": [...] }` or `{ "success": false, "error": "..." }` with 400/403/404/409. If Inventory is built first, both modules may share one dispatcher with a `MODULE` column on the registry (open decision D8).

Procedure signature convention: `IN p_xxx` (strings, numbers, ISO dates; complex payloads as `p_lines_json` `CLOB`), `IN p_user`, `OUT p_id` `NUMBER`, `OUT p_number` `VARCHAR2`, `OUT p_status` `VARCHAR2` ('S'/'W'/'E'), `OUT p_message` `VARCHAR2`.

9.3 Package specifications

Package	Procedures (all with <code>p_user</code> and the OUT set)	Key validations (section 12)
<code>RR_PO_SETUP_PKG</code>	<code>SAVE_BU_OPTIONS</code> , <code>SAVE_LOCATION</code> , <code>SAVE_CATEGORY</code> , <code>SAVE_EXPENSE_ITEM</code> , <code>SAVE_BUYER</code> , <code>SAVE_REQUESTER_DEFAULT</code> , <code>SAVE_ACCOUNT_RULE</code> , <code>SAVE_DOC_SEQUENCE</code> , <code>SAVE_SITE_OPTIONS</code> , <code>SET_STATUS(p_entity, p_id, p_status)</code>	S-01...S-12

Package	Procedures (all with <code>p_user</code> and the OUT set)	Key validations (section 12)
RR_PO_ACCOUNT_PKG	functions <code>DERIVE_CHARGE_ACCOUNT(bu, requester, category, item)</code> , <code>VALIDATE_ACCOUNT(combo, bu)</code> , <code>NEXT_DOC_NUMBER(bu, doc_type)</code>	S-08, REQ-08
RR_PO_REQ_PKG	<code>SAVE_REQUISITION(p_header_json, p_lines_json)</code> (create/update draft incl. distributions), <code>DELETE_DRAFT</code> , <code>SUBMIT</code> , <code>WITHDRAW</code> , <code>CANCEL(p_req_header_id, p_line_ids)</code> , <code>RETURN_LINES(p_line_ids, p_reason)</code> (buyer → requester), <code>COPY_REQUISITION</code>	REQ-01... REQ-15
RR_PO_DOC_PKG	<code>CREATE_PO(p_header_json, p_lines_json)</code> (direct PO, 6.2.0), <code>COPY_PO(p_po_header_id)</code> , <code>UPDATE_PO</code> (INCOMPLETE/REJECTED only), <code>DELETE_PO</code> (never approved), <code>AUTOCREATE(p_req_line_ids_json, p_options_json)</code> , <code>SUBMIT</code> , <code>WITHDRAW</code> , <code>HOLD/RELEASE_HOLD</code> , <code>CANCEL(p_level, p_id, p_reason, p_recreate_demand)</code> , <code>CLOSE/REOPEN(p_level, p_id)</code> , <code>FINAL_CLOSE(p_po_header_id)</code> , <code>COMMUNICATE(p_po_header_id, p_attachment_id, p_to, p_cc)</code> , <code>RECORD_ACCEPTANCE</code>	PO-01...PO-20
RR_PO_CHANGE_PKG	<code>SAVE_CHANGE_ORDER(p_po_header_id, p_changes_json, p_reason)</code> , <code>SUBMIT</code> , <code>CANCEL</code> , <code>APPLY</code> (internal, from approval callback)	CO-01... CO-09
RR_PO_RCV_PKG	<code>RECEIVE(p_header_json, p_lines_json)</code> (many schedules, one delivery note), <code>RETURN_TO_SUPPLIER(p_rcv_transaction_id, p_qty_or_amount, p_reason, p_date)</code> , <code>CORRECT(p_rcv_transaction_id, p_delta, p_date)</code> ; function <code>CLOSURE_ROLLUP(p_schedule_id)</code>	RCV-01... RCV-12
RR_PO_MATCH_PKG	<code>MATCH_TO_PO(p_invoice_id, p_matches_json)</code> , <code>MATCH_TO_RECEIPT(p_invoice_id, p_matches_json)</code> , <code>REVERSE_MATCH(p_match_id)</code> , <code>VALIDATE_INVOICE(p_invoice_id)</code> , <code>RELEASE_HOLD(p_hold_id, p_release_code, p_reason)</code> , <code>PLACE_MANUAL_HOLD</code>	MAT-01... MAT-14
RR_PO_ACCT_PKG	<code>CREATE_RECEIPT_ACCOUNTING(p_bu, p_to_date)</code> , <code>RUN_PERIOD_END_ACCRUAL(p_bu, p_period)</code> , <code>POST_ACCRUAL_RUN(p_run_id)</code> , <code>CANCEL_ACCRUAL_RUN</code> , <code>WRITE_OFF_ACCRUALS(p_distribution_ids_json, p_reason, p_date)</code>	ACC-01... ACC-10
RR_PO_APPROVAL_PKG	<code>REQUEST(p_module, p_type, p_id)</code> (internal), <code>ON_DECISION(p_request_id, p_transaction_id, p_decision, p_user)</code> (approval callback)	APR-01... APR-05
RR_PO_BUDGET_PKG	Phase 1 stubs <code>CHECK_FUNDS</code> , <code>RESERVE_FUNDS</code> , <code>RELIEVE_FUNDS</code> returning <code>PASSED</code>	Phase 3

9.4 Frontend service layer

```
// src/services/po.service.ts – the only two calls the module uses
export const poQuery = (sql: string) =>
  fetch(`${base}/ai/executequery`, { method: 'POST',
    body: JSON.stringify({ sql, maxRows: 2000, appUser: 'PROCUREMENT' }) });

export const poExec = (procedure: string, params: Record<string, unknown>) =>
  fetch(`${base}/po/execute`, { method: 'POST',
    body: JSON.stringify({ procedure, params, appUser: currentUser }) });

// examples
poExec('RR_PO_REQ_PKG.SUBMIT', { p_req_header_id: 6000000123 });
poExec('RR_PO_RCV_PKG.RECEIVE', {
  p_header_json: JSON.stringify({ supplierSiteId: 300000045, receiptDate: '2026-10-05', deliveryNote: 'DN-88',
  p_lines_json: JSON.stringify([ { scheduleId: 6000000201, quantity: 20 } ]) });
```

10. Navigation and Screens

10.1 Menu (module code **po**, label "Purchasing")

The native module lives under `/po/*` so it does not collide with the existing Fusion-based `/procurement/*` screens, which stay available for Fusion users (open decision D9).

Group	Menu entry	Path
Requisitions	My Requisitions	<code>/po/requisitions</code>
	Shop / Create Requisition (catalog + free text)	<code>/po/requisitions/new</code>
Purchasing	Process Requisitions (buyer workbench)	<code>/po/workbench</code>
	Purchase Orders	<code>/po/orders</code>
	Create Purchase Order	<code>/po/orders/new</code>
	Change Orders	<code>/po/change-orders</code>
Receiving	Receive	<code>/po/receive</code>
	Receipts (returns and corrections)	<code>/po/receipts</code>
Invoicing	Match Invoice to PO (opens AP Create Invoice in match mode)	<code>/ap/manage-invoices?match=po</code>
	Invoice Holds	<code>/ap/invoice-holds</code>
Accruals	Uninvoiced Receipts & GRNI Reconciliation	<code>/po/accruals</code>
	Period-End Accrual	<code>/po/period-end-accrual</code>
Reports	Open POs, PO Status (3-way), Spend Analysis, Requisition Aging, Approval Turnaround, Supplier Delivery Performance	<code>/po/reports/*</code>
Setup	Purchasing Options (per BU), Locations, Categories, Expense Item Catalog, Buyers, Requester Defaults, Account Rules, Document Numbering, Supplier Site Options, Approval Rules (existing Approvals page filtered to PROCUREMENT)	<code>/po/setup/*</code>

10.2 Screens overview

Screen	Pattern (consistent with existing Re-ERP pages)
Purchasing landing	KPI cards (open requisitions, pending approvals, open PO value, GRNI balance, invoices on hold, overdue deliveries) + menu tiles
My Requisitions	Search + grid with status tags; tabs per opened requisition (same multi-tab pattern as Manage Payments)
Create Requisition	Catalog cards with search by category, "Add to requisition", free-text line form; right panel = cart with derived account per line; Submit
Buyer workbench	Grid of approved lines grouped by suggested supplier; select → Autocreate dialog (supplier, site, currency, combine lines)
PO form	Header form + lines grid with expandable schedules/distributions; quantity-ledger columns (ordered / received / billed / cancelled) read-only after approval; Actions menu: Submit, Change, Cancel, Close, Final close, Communicate, Print PDF, View revisions
Receive	Search open schedules → editable "Receive now" column → one receipt; shows over-receipt warnings inline
Match Invoice (AP)	In Create Invoice: "Match to PO" switch → pick schedules or receipts → qty/price grid with live IPV/ERV and tolerance status → creates invoice lines
Invoice holds	Grid of held invoices by hold code with release dialog (release code + reason)
Accruals	Uninvoiced receipts grid; GRNI recon card (sub-ledger vs GL); period-end accrual draft → post

Every screen keeps the existing API-transparency button showing the exact po/execute call and response.

10.3 Key wireframes

Create Requisition

[Save draft] [Submit]

Category ▾ [IT ▶ Accessories]

Search [toner]

CART (BU: DIFC)

1 HP 26A toner ×4 1,120.00

01-10-200-6106002-...

2 Consultancy (AMOUNT)

15,000.00 split 60/40

Need by 15-Oct Deliver HQ

Total AED 16,120.00

HP 26A

280 AED/EA

[Add]

USB-C dock

390 AED/EA

[Add]

Headset

150 AED/EA

[Add]

[+ Free-text item]

PO PO-DIFC-2026-00017 Rev 1 APPROVED · OPEN

[Actions ▾] [PDF] [Send] [API]

Supplier Gulf Office Supplies LLC — Site DXB-MAIN

Buyer Aisha K. AED · Net 30

Line	Item	UOM	Ordered	Price	Received	Billed	Cancel	Status
1	Office chair ergonomic	EA	20	450.00	18	18	2	CLOSED
	▶ Sch 1 HQ need-by 10-Oct 3-way accrue@receipt ▶ Dist 1 01-10-200-6105001-... ordered 20 delivered 18 billed 18 GRNI 0.00							
2	AC maintenance (svc)	-	15,000.0	-	-	3,000.00	-	OPEN

Receive

Receipt date [05-Oct-2026] Delivery note [DN-8812]

PO	Line	Item	Ordered	Received	Receive now	Deliver to
PO-...-00017	1	Office chair ergonomic	20	0	[20]	HQ Floor 3
PO-...-00019	1	Printer paper A4	100 BX	60	[40]	HQ Store

[Cancel] [Receive]

11. Reports

Report	Content	Source
Open purchase orders	Open value by supplier, buyer, category, need-by aging	RR_PO_V_ORDERS
PO status (3-way)	Per schedule: ordered, received, billed, cancelled, open; holds	RR_PO_V_ORDER_LINES + matches
Uninvoiced receipts / GRNI	Accrued not billed by supplier, age buckets; GL tie-out	RR_PO_V_ACCRUAL_RECON
Spend analysis	Spend by category, supplier, cost centre, month; PO vs non-PO invoice spend (compliance)	RR_PO_V_SPEND + AP invoices without PO
Requisition aging	Approved lines not yet on a PO, by buyer, days waiting	RR_PO_V_REQ_LINES
Approval turnaround	Submit → approval time by approver and document type	RR_APPROVAL_HISTORY
Supplier delivery performance	On-time receipt %, over/under receipts, returns %	receipts vs need-by / promised
Price variance	IPV by supplier and category	RR_PO_V_MATCHES
Commitments (Phase 3)	Budget vs pre-commitment vs commitment vs actual	funds ledger

All reports export to Excel and PDF with the existing helpers; the AI assistant can query the `RR_PO_V_*` views through the same read gateway.

12. Validation Rules Catalog

Every write validates payload and state before touching data and returns 400 (bad payload), 403 (no access), 404 (record missing) or 409 (business rule) with a specific message the screen shows verbatim. Screens repeat the cheap checks client-side; the server is authoritative.

12.1 Setup (S)

Code	Rule
S-01	One <code>RR_PO_BU_OPTIONS</code> row per BU; BU exists and is active in <code>RR_GL_BUSINESS_UNITS</code>
S-02	Accrual, exchange gain/loss and (when set) IPV and write-off accounts are valid combinations of the BU's company
S-03	Tolerances between 0 and 100; <code>OVER_RECEIPT_ACTION</code> in (<code>REJECT</code> , <code>WARNING</code>)
S-04	Location code unique; at least one of ship-to / bill-to / deliver-to flags
S-05	Category code unique; no cycles in the tree; natural account value exists in the account value set
S-06	Expense item code unique; quantity items need a UOM; category active and requestable
S-07	Buyer user exists with <code>P0</code> module access; one default buyer per BU
S-08	Requester template account valid; segment 1 = BU company
S-09	Account rule segment number 1–10; value exists in that segment's value set
S-10	One sequence per BU and doc type; <code>NEXT_NUMBER</code> ≥ 1
S-11	Supplier site option: site exists with <code>PURCHASING_FLAG</code> = <code>Y</code> ; email valid when method <code>EMAIL</code>
S-12	Inactivating a category/item/location/buyer used by open documents is allowed (no new use) but warned

12.2 Requisitions (REQ)

Code	Rule
REQ-01	User has <code>P0</code> module access and access to the requisition BU
REQ-02	Only <code>INCOMPLETE</code> / <code>REJECTED</code> requisitions can be edited; only never-submitted drafts deleted
REQ-03	At least one line; line numbers unique
REQ-04	Category active and requestable; catalog item active
REQ-05	Quantity line: UOM valid, quantity > 0 , price ≥ 0 ; amount line: amount > 0
REQ-06	Need-by date $\geq$ submit date
REQ-07	Deliver-to location active and usable by the BU
REQ-08	Distribution percentages sum to 100 per line; every charge account valid for the BU company
REQ-09	Foreign currency requires a rate on the rate date
REQ-10	Suggested supplier site (when given) is a purchasing site of the BU
REQ-11	Submit requires an approval rule unless approval not required
REQ-12	Withdraw only while <code>PENDING_APPROVAL</code> ; cancels the approval request
REQ-13	Cancel only lines not <code>ON_PO</code>
REQ-14	Return-to-requester only by a buyer, only <code>OPEN</code> approved lines, reason required
REQ-15	Requester (on behalf) must have access to the BU

12.3 Purchase orders (PO)

Code	Rule
PO-01	Buyer active for the BU
PO-02	Supplier active (RR_SUPPLIER_MASTER); site active, PURCHASING_FLAG = Y , assigned to the PO BU (RR_SUPPLIER_SITE_ASSIGNMENTS.CLIENT_BU_ID or PROCUREMENT_BU_ID), no purchasing hold (4.10)
PO-03	Currency rate available; payment terms present
PO-04	Ship-to / bill-to locations valid for the BU
PO-05	Each line has ≥ 1 schedule; each schedule ≥ 1 distribution; quantity ledger sums consistent
PO-06	Quantity line: UOM, quantity > 0, price ≥ 0 ; amount line: amount > 0
PO-07	Charge accounts valid for the BU company; tax codes active and assigned to the BU
PO-08	Direct POs allowed by default; only when the BU sets REQUIRE_REQUISITION = Y must a manual PO come from a buyer with DIRECT_PO_ALLOWED = Y
PO-09	Requisition lines on autocreate are APPROVED / OPEN , same BU, not already on a PO (row lock)
PO-10	Autocreate grouping keys equal within one PO
PO-11	Edit only INCOMPLETE / REJECTED ; approved POs change only through change orders
PO-12	Delete only POs never approved; linked requisition lines return to OPEN
PO-13	Submitter $\neq$ sole approver (SoD)
PO-14	Cancel: amount to cancel = ordered – max(delivered, billed) > 0
PO-15	Close/reopen: not FINALLY_CLOSED ; reopen recomputes the automatic status
PO-16	Final close: no unreleased holds on matched invoices; receipt-accrual distributions have ACCRUED_AMOUNT_FUNC = 0
PO-17	Hold: approved PO; while on hold no receiving or matching
PO-18	Communicate: approved; email address present for EMAIL
PO-19	After-the-fact PO only when ALLOW_AFTER_FACT_PO = Y ; flagged
PO-20	PO total and BU functional amounts recomputed server-side; client totals ignored

12.4 Change orders (CO)

Code	Rule
CO-01	PO APPROVED , not FINALLY_CLOSED , not on hold; one open change order per PO

Code	Rule
CO-02	FROM_REVISION = current revision at submit and at apply
CO-03	New quantity/amount $\geq$ max(delivered, billed)
CO-04	Price change only when nothing billed on the line
CO-05	Account change only when nothing delivered or billed on the distribution
CO-06	Supplier, site, currency cannot change
CO-07	Re-approval when the increase exceeds CO_REAPPROVAL_THRESHOLD_PCT or a line is added
CO-08	Apply re-runs PO-04...PO-07
CO-09	Apply writes a revision and re-communicates when the site method is EMAIL

12.5 Receiving (RCV)

Code	Rule
RCV-01	User may receive (receiver role, requester of the line, or buyer when SoD allows)
RCV-02	All lines of one receipt belong to one supplier site and one BU
RCV-03	PO approved, not on hold; schedule OPEN or CLOSED_FOR_INVOICING, not cancelled
RCV-04	Receipt date $\leq$ today, $\geq$ PO approval date, in an open GL period
RCV-05	Early receipt check when EARLY_RECEIPT_DAYS set
RCV-06	Quantity lines: quantity > 0 in the PO UOM (UOM conversion Phase 2b); amount lines: amount > 0
RCV-07	Over-receipt tolerance (REJECT error / WARNING warning)
RCV-08	Return $\leq$ received – returned – billed against that receipt
RCV-09	Return reason required
RCV-10	Negative correction same limit as return; positive correction obeys RCV-07
RCV-11	No receiving on a schedule with a pending change order that reduces it below the new total
RCV-12	Delivery note number unique per supplier site (warning)

12.6 Matching (MAT)

Code	Rule
MAT-01	Invoice supplier = PO supplier; invoice site = PO site or another pay site of the same supplier
MAT-02	Invoice currency = PO currency
MAT-03	PO approved, not on hold; schedule OPEN or CLOSED_FOR_RECEIVING
MAT-04	3-way schedules match receipts (quantity not yet received may be matched ahead and is held); 2-way schedules match the PO
MAT-05	Billed beyond received (3-way) or beyond ordered (any) is never an error: it places QTY_REC / QTY_ORD holds that release when the condition clears
MAT-06	Unit price > 0 ; variance beyond tolerance $\rightarrow$ PRICE hold (not an error)
MAT-07	Credit memo match: negative quantity $\leq$ billed

Code	Rule
MAT-08	Match rows pro-rated to distributions; rounding to the last distribution
MAT-09	Accounting date in an open AP and GL period
MAT-10	Reverse match only when the invoice is not paid and not accounted (else credit memo)
MAT-11	Validation places/releases holds idempotently (re-running gives the same result)
MAT-12	Manual release requires a release code and reason; releaser $\neq$ user who matched (price holds)
MAT-13	Invoices with unreleased holds are excluded from accounting and payment selection
MAT-14	Matching recomputes closure for every affected schedule

12.7 Accounting and accruals (ACC)

Code	Rule
ACC-01	Create Accounting only for UNACCOUNTED receiving transactions in open GL periods
ACC-02	Every generated journal balances (functional amounts)
ACC-03	Period-end accrual: GL period open; one posted run per BU/period
ACC-04	Accrual uses as-of-period-end delivered and billed quantities
ACC-05	Draft run can be regenerated; posted run cancellable only while the period is open
ACC-06	Reversal dated first day of next period, which must exist
ACC-07	Write-off $\leq$ accrued balance; reason required; only closed distributions or older than the BU threshold
ACC-08	Write-off account valid for the BU company
ACC-09	GRNI recon difference reported, never auto-adjusted
ACC-10	REFERENCE2/REFERENCE5 set on every journal line

12.8 Approvals (APR)

Code	Rule
APR-01	Rule selection: module, type, amount band, BU, category, priority
APR-02	Submitter skipped as approver
APR-03	Callback moves the document only from PENDING_APPROVAL (idempotent on repeat calls)
APR-04	Withdraw cancels the open request; resubmit starts a new request
APR-05	Approver acting by email token gets the same validations as in-app approval

13. Delivery: Scripts, Seeding, Migration

- Scripts in `database/po/3xx_*.sql`, one per sub-phase, each with a verification query in its footer (same style as `database/gl`, `database/ap`). ORDS handler source uses `q'~ ... ~'` quoting.

- Seeding: UOMs; categories IT, IT.HARDWARE, IT.SOFTWARE, IT.ACCESSORIES, OFFICE.STATIONERY, OFFICE.FURNITURE, FACILITIES.MAINTENANCE, FACILITIES.CLEANING, PROF.CONSULTING, PROF.LEGAL, PROF.AUDIT, MKT.EVENTS, MKT.ADVERTISING, TRAVEL, RENT (natural accounts left blank for finance to fill); document sequences per active BU; approval callbacks; registry rows for every procedure.
- Migration (optional): open Fusion POs can be imported with ORIGIN = IMPORT, LEGACY_PO_NUMBER set, ordered / received / billed quantities taken from Fusion, accrual balances seeded by a one-off journal so the GRNI reconciliation starts at zero difference.
- Feature flags: the menu shows the module only to users with PO module access; AP matching UI appears only when the BU has RR_PO_BU_OPTIONS.

14. Delivery Roadmap

Phase	Content	Depends on
1a Setup	Tables, sequence, po/execute dispatcher + registry + log, setup packages and pages, read views	-
1b Requisitions	Requisition pages, account derivation, approval engine extensions (rule filters, callbacks), attachments, action history	1a
1c Purchase orders	PO pages, buyer workbench, autocreate, approval, revisions, PDF and email	1b
1d Receiving	Receive, returns, corrections, receipt accounting (SLA → GL)	1c
1e Matching	AP match mode, matches, holds, variances, payment/accounting exclusions	1d
1f Accruals	Period-end accrual, GRNI reconciliation, write-off, reports	1e
1g Change & close	Change orders, cancel, close, final close	1c (1e for billed checks)
2 Agreements	Blanket agreements and releases, sourcing requisitions from agreements	1g
2b Inventory lines	INVENTORY line type, receipt into the Inventory engine (PO_RECEIPT)	Inventory 1c
3 Budgetary control	Budgets, funds check, encumbrance accounting	1f
4 Sourcing & portal	RFQs, quotations, supplier portal acknowledgements	2

Each sub-phase ships as one database script plus its pages and is deployable independently.

15. Open Decisions

#	Decision	Recommendation
D1	Default accrual method	Receipt accrual for quantity lines, period-end for amount (service) lines
D2	Default match level	3-way for quantity lines, 2-way for services
D3	Approval routing	Amount-band rules per BU/category (existing engine) now; supervisor chain later via MANAGER_USER_NAME
D4	Requisitions mandatory?	Decided (v1.1): no — POs are created directly by default; requisitions optional per BU
D5	Tolerances	Over-receipt 0 % reject; invoice qty 0 %; invoice price 5 % or 500 AED
D6	ERV on period-end-accrual POs	Separate ERV (keeps expense at PO value)
D7	Non-recoverable VAT	Not needed now (all input VAT recoverable); add a recovery rate on tax codes if required

#	Decision	Recommendation
D8	Shared dispatcher with Inventory	Share one dispatcher with a module column if Inventory is built in the same release
D9	Existing Fusion /procurement/* screens	Keep during parallel run; hide per company once POs are native
D10	Supplier master	Decided (v1.1): existing RR_SUPPLIER_* tables only (kept by the current Fusion sync); no Procurement supplier master. Re-ERP supplier creation only if Fusion is retired

16. Glossary

Term	Meaning
GRNI	Goods received not invoiced — the receipt accrual liability
IPV	Invoice price variance — invoice price differs from PO price
ERV	Exchange rate variance — invoice rate differs from the PO/receipt rate
2-way match	Invoice checked against the PO only
3-way match	Invoice checked against the PO and the receipt
Schedule	A delivery of a PO line (date, location, quantity) — also called a shipment or line location
Distribution	The account split of a schedule; holds the quantity ledger
Quantity ledger	Ordered, delivered, billed and cancelled counters that every event updates
Change order	An approved, versioned modification of an approved PO

Appendix A — Executable simulation

`docs/design/po_rd_simulation.py` implements the quantity ledger, receiving, 2-way/3-way matching with holds, variances, returns, period-end accrual and closure exactly as specified in sections 5–7, runs worked examples A–E and asserts the invariants (every journal balances, GRNI clears to zero after full matching, `invoice = base + IPV + ERV`, closure statuses as expected, over-receipt and over-billing rejected or held). Run:

```
python3 docs/design/po_rd_simulation.py
```

The output of the last run is reproduced below.

A. Office chairs - quantity line, 3-way, receipt accrual
 [PASS] over-receipt 21 of 20 rejected - RCV-07 over-receipt: 21 > 20
 [PASS] receipt accrual Dr expense / Cr GRNI 9,000.00
 [PASS] invoice 20 x 460 + VAT = 9,660.00 - liability 9660.00
 [PASS] IPV 200.00 to expense, ERV 0
 [PASS] price +2.2% within 5% tolerance - no hold - []
 [PASS] GRNI cleared to 0.00
 [PASS] expense = 9,200.00 (actual price)
 [PASS] schedule CLOSED - CLOSED
 [PASS] all journals balanced

A2. Holds - over-billing and price variance
 [PASS] billed 12 > received 10 -> QTY_REC hold - ['QTY_REC', 'PRICE']
 [PASS] 480 vs 450 (+6.7%) > 5% -> PRICE hold - ['QTY_REC', 'PRICE']
 [PASS] after receiving 2 more, QTY_REC releases
 [PASS] 12 received = 12 billed -> CLOSED_FOR_INVOICING (3-way) - CLOSED_FOR_INVOICING
 [PASS] receiving 3 more re-opens it for invoicing -> OPEN - OPEN
 [PASS] all journals balanced

B. AC maintenance - amount line, 2-way, quarterly schedules
 [PASS] Q1 invoice 3,000 + VAT 150 = 3,150.00 - 3150.00
 [PASS] expense 3,000.00 (no accrual account used)
 [PASS] Q1 schedule CLOSED by billing (2-way) - CLOSED
 [PASS] Q2-Q4 still OPEN
 [PASS] all journals balanced

C. Software licences - USD PO, receipt accrual, exchange variance
 [PASS] receipt at PO rate: 1,200 x 3.6725 = 4,407.00
 [PASS] ERV loss 9.00 - 9.00
 [PASS] VAT 60 USD x 3.68 = 220.80 - 220.80
 [PASS] liability 1,260 USD x 3.68 = 4,636.80 - 4636.80
 [PASS] GRNI cleared
 [PASS] all journals balanced

D. Return before invoicing, then cancel the remainder
 [PASS] return 2: GRNI 8,100.00
 [PASS] return beyond received-not-billed rejected - RCV-08 return 19 > received-not-billed 18
 [PASS] invoice 18 clears GRNI
 [PASS] CLOSED_FOR_INVOICING (18 of 20 received) - CLOSED_FOR_INVOICING
 [PASS] cancel remaining 2 -> CLOSED - CLOSED
 [PASS] all journals balanced

E. Stationery - period-end accrual
 [PASS] receipt creates no journal (accrue at receipt = N)
 [PASS] 30 Sep accrual 500.00
 [PASS] Sep expense 500.00 via accrual
 [PASS] accrued liabilities back to 0.00 after reversal
 [PASS] Oct expense net 0.00 (reversal -500 + invoice +500)
 [PASS] second run on 31 Oct accrues nothing (fully billed)
 [PASS] all journals balanced

All scenarios passed.